南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

生物医学工程系
Department of Biomedical Engineering

Frequency Filtering

Yiming Li
liym2019@sustech.edu.cn

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Lecture Roadmap

Implement filtering without wraparound

Smooth: compare blur and ringing

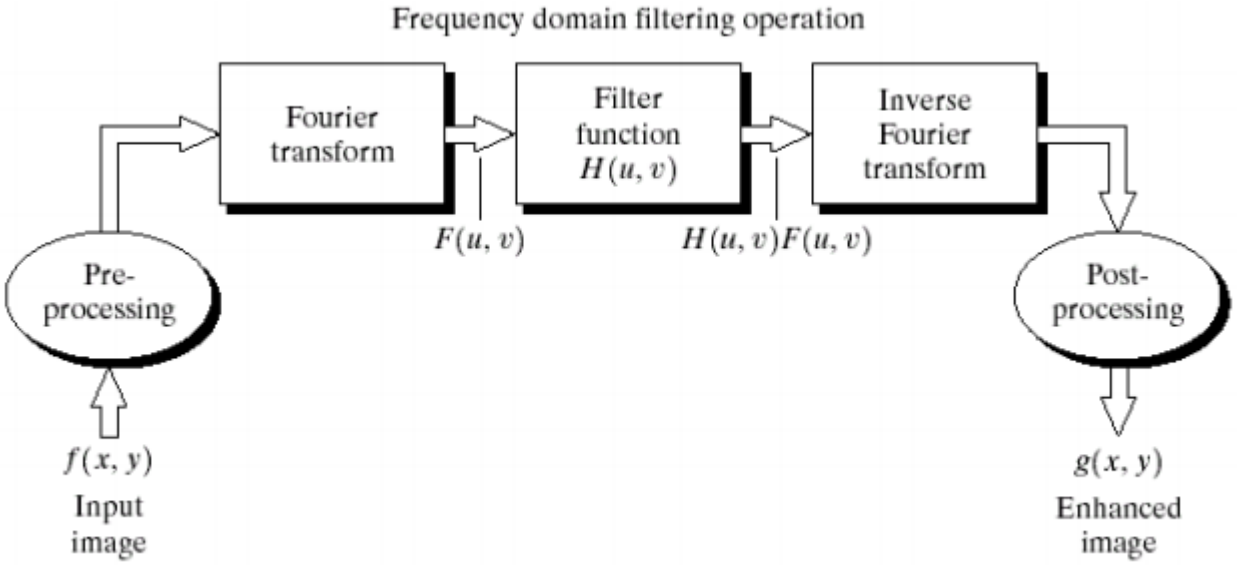
Sharpen: add detail while retaining brightness

Correct illumination and target periodic interference

The DFT and Image Processing

- To filter an image in the frequency domain:

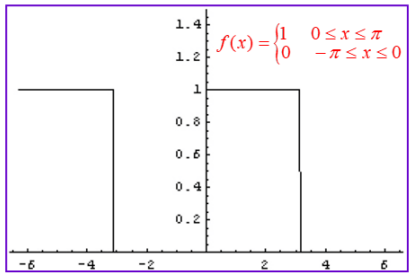
1. Compute *F(u,v)* the DFT of the image
2. Multiply *F(u,v)* by a filter function *H(u,v)*
3. Compute the inverse DFT of the result

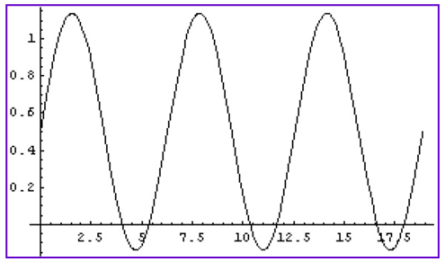


The DFT and Image Processing

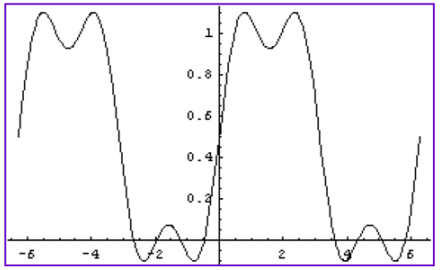
- Low frequencies in the transform are related to slowly varying intensity components. High frequencies are caused by sharp transitions in intensity, such as edges and noise.

例如:方波的傅立叶表示

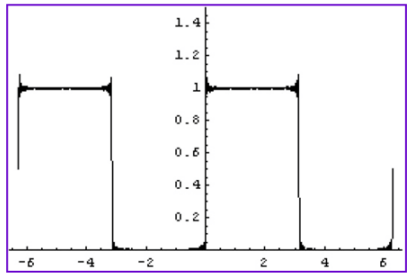




$$\frac{1}{2}+\frac{2}{\pi}\sin(x)$$

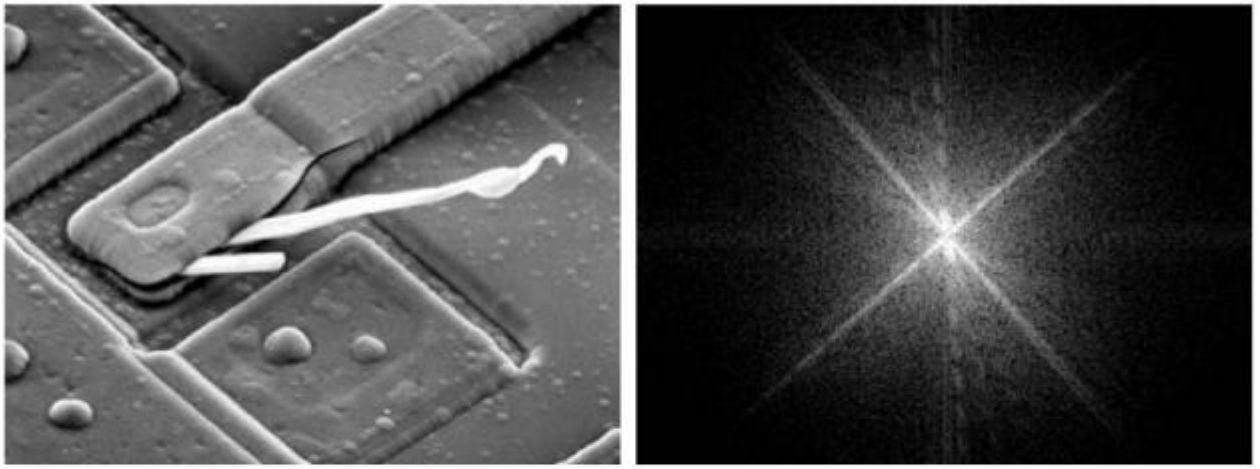


$$\frac{1}{2}+\frac{2}{\pi}\sin(x)+\frac{2}{3\pi}\sin(3x)$$



$$\frac{1}{2}+\frac{2}{\pi}\sum_{n=0}^{49}\sin\left((2n+1)x\right)$$

Some Basic Frequency Domain Filters

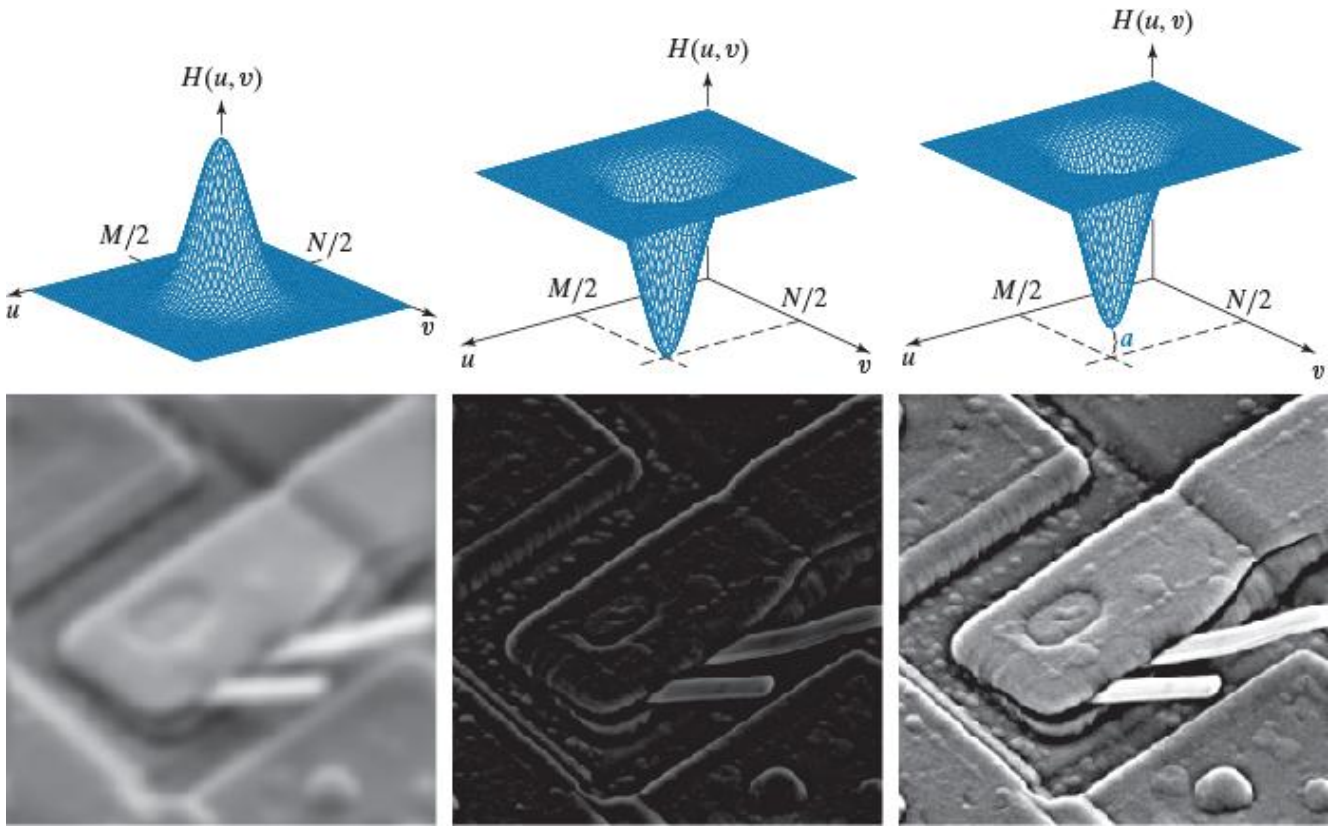


a b

FIGURE 4.29 (a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

The DFT is centered after multiplication of the image by $(-1)^{m+n}$

Some Basic Frequency Domain Filters

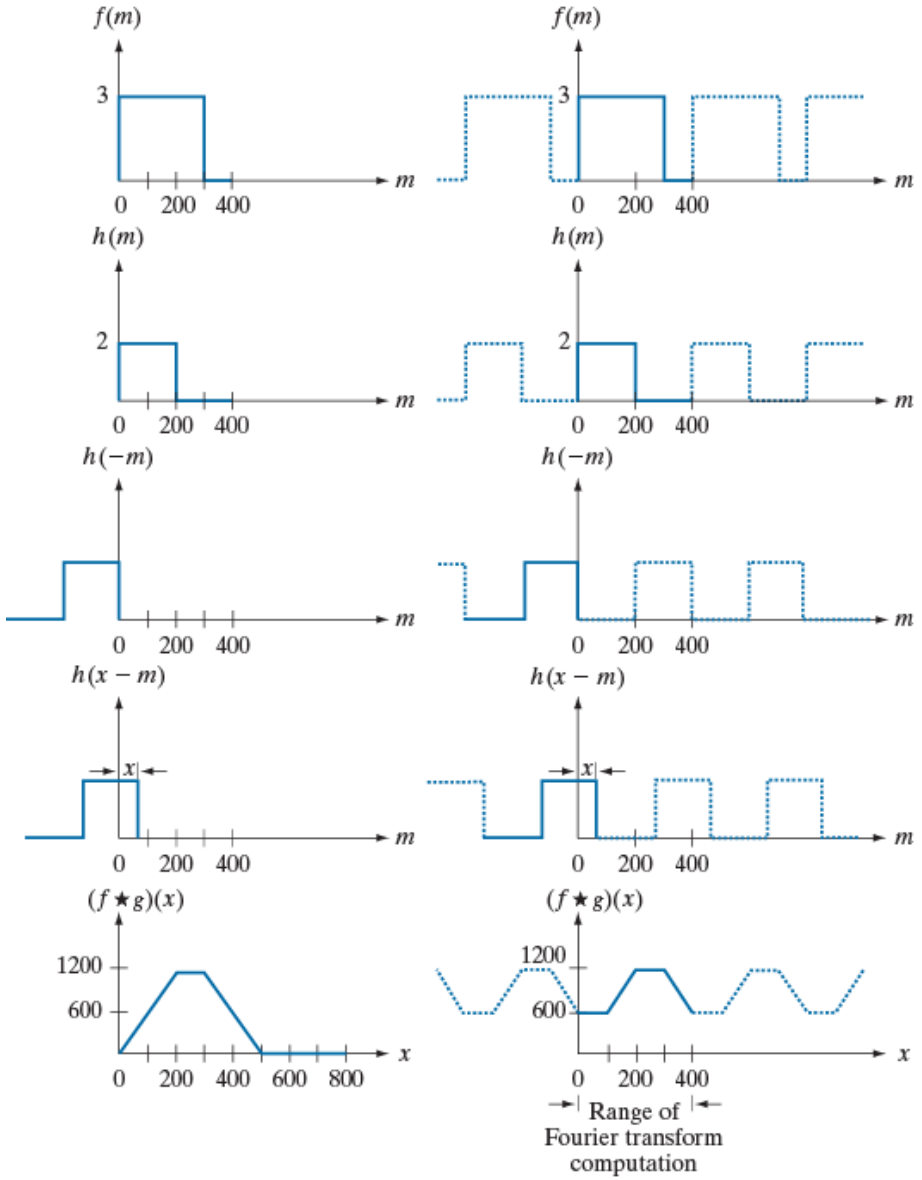


a b c
d e f

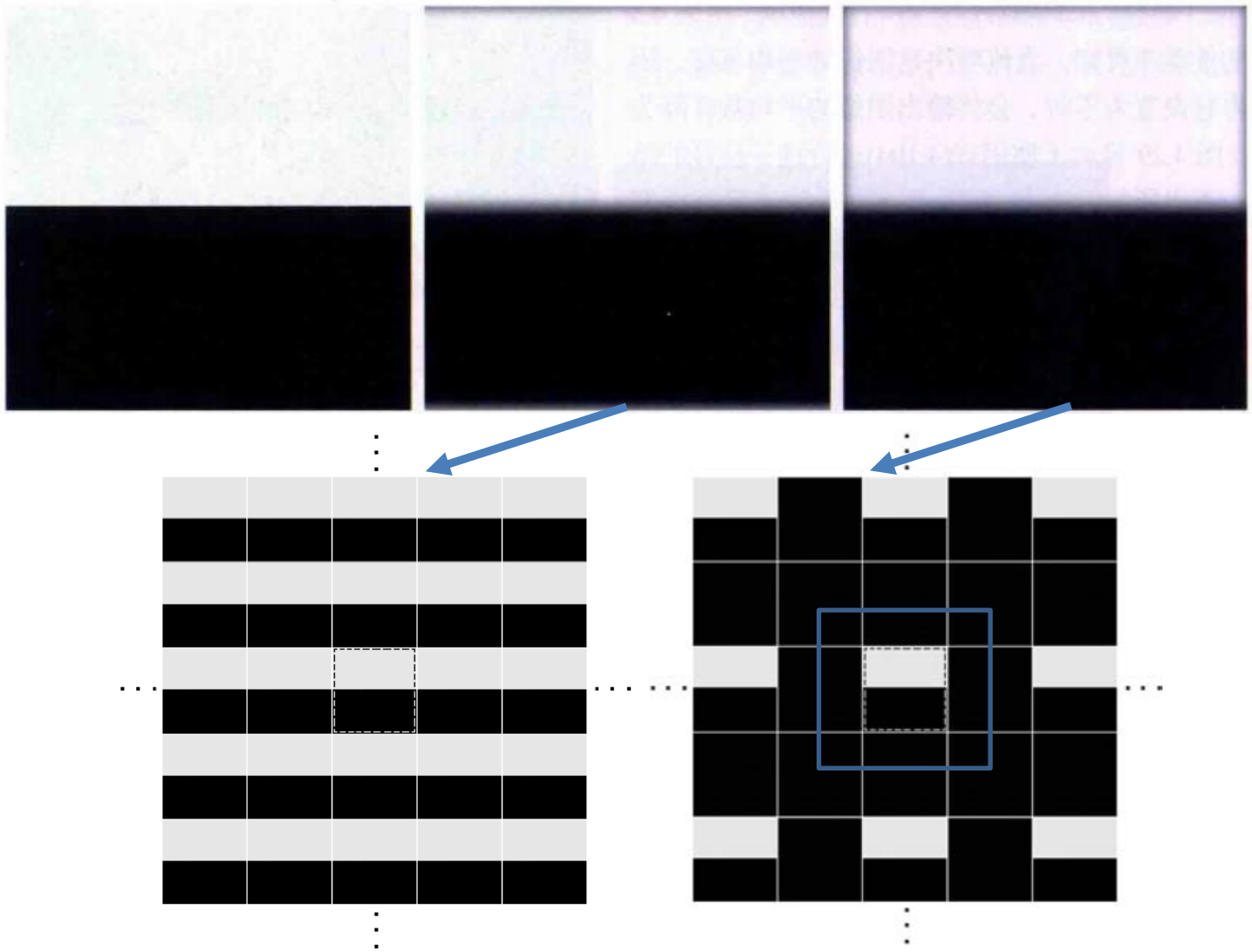
FIGURE 4.30 Top row: Frequency domain filter transfer functions of (a) a lowpass filter, (b) a highpass filter, and (c) an offset highpass filter. Bottom row: Corresponding filtered images obtained using Eq. (4-104). The offset in (c) is $a = 0.85$, and the height of $H(u,v)$ is 1. Compare (f) with Fig. 4.28(a).

The importance of zero padding

The DFT considers that the signal is periodic and produces wraparound errors.



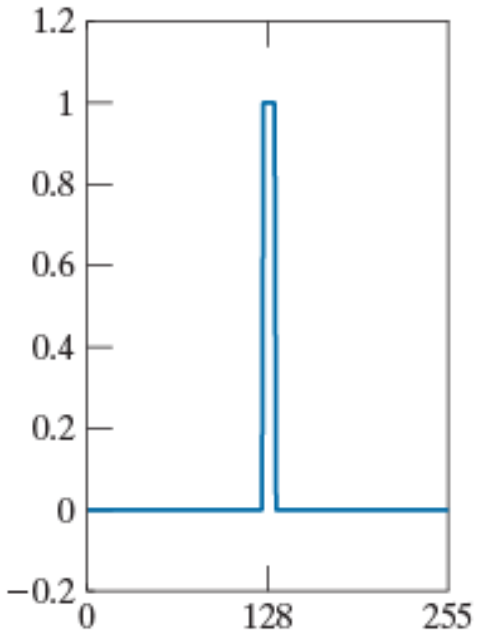
The importance of zero padding

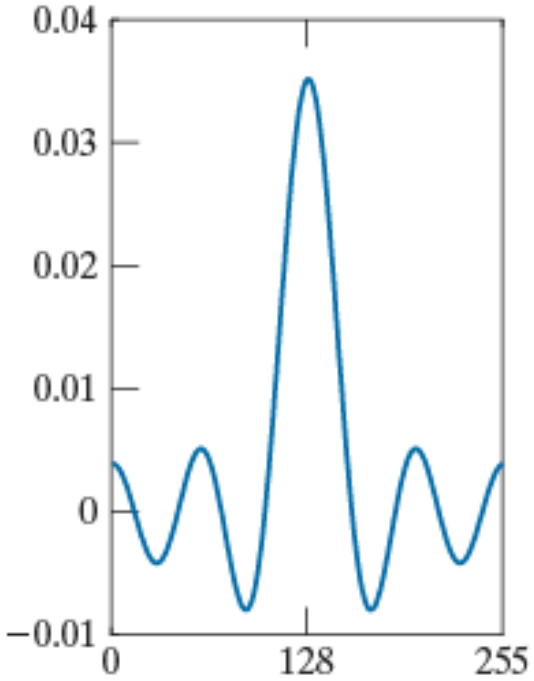


$$g(x,y) = \mathcal{F}^{-1}[H(u,v)F(u,v)]$$

Spatial zero-padding and filters

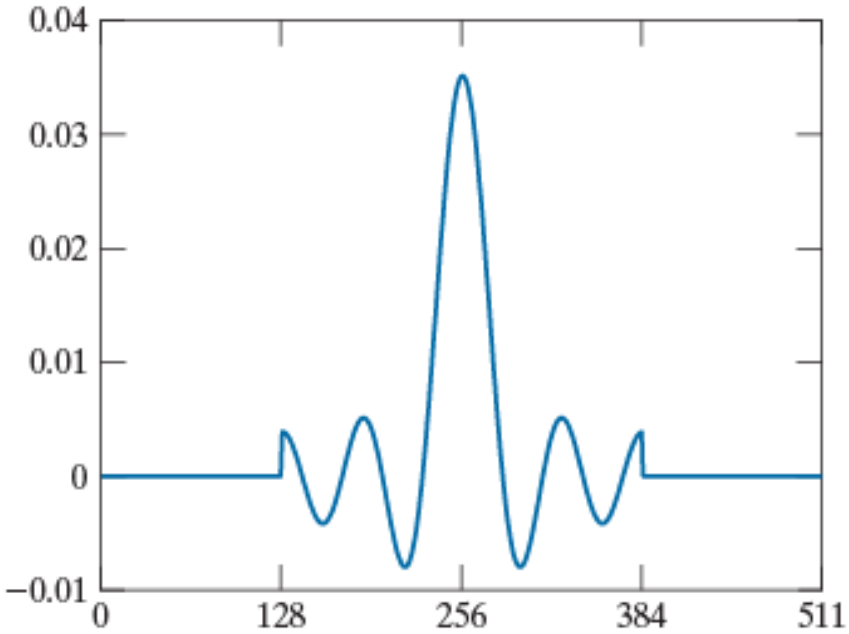
The filter and its inverse DFT of length 256 (continuous line)

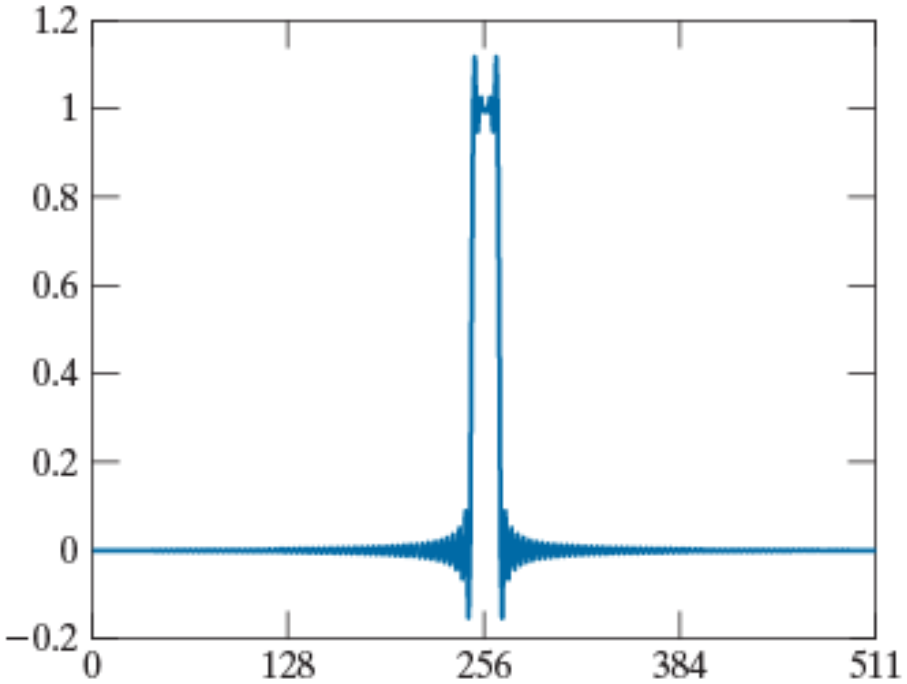




Spatial zero-padding and filters

Zero-padded filter and its DFT





Spatial truncation of the filter results in ringing effects.

Spatial zero-padding and filters

- We cannot work with an infinite number of filter components and simultaneously perform zero-padding to avoid aliasing.
- A decision on which limitation to accept is required.
- One solution to zero-pad the image and then use a filter of the same size with no zero-padding
  - Small errors due to aliasing but it is generally preferable than ringing.
- Another solution is to choose filters attenuating gradually instead of ideal filters.

Steps of filtering in the DFT domain

1. 对给定大小为$M \times N$的图像$f(x,y)$，选择合适的填充参数P和Q，$P = 2M, Q = 2N$
2. 对$f(x,y)$添加必要数量的0，形成大小为$P \times Q$的图像$f_p(x,y)$
3. 用$(-1)^{x+y}$乘以$f_p(x,y)$移到其变换的中心
4. 计算其DFT，得到$F(u,v)$
5. 设计滤波函数$H(u,v)$,其大小为$P \times Q$,中心在$(\frac{P}{2}, \frac{Q}{2})$处。用阵列相乘得到$G(u,v) = H(u,v)F(u,v)$
6. 计算IDFT，为忽略由于计算不准确导致的寄生复分量，只选择计算后的实部

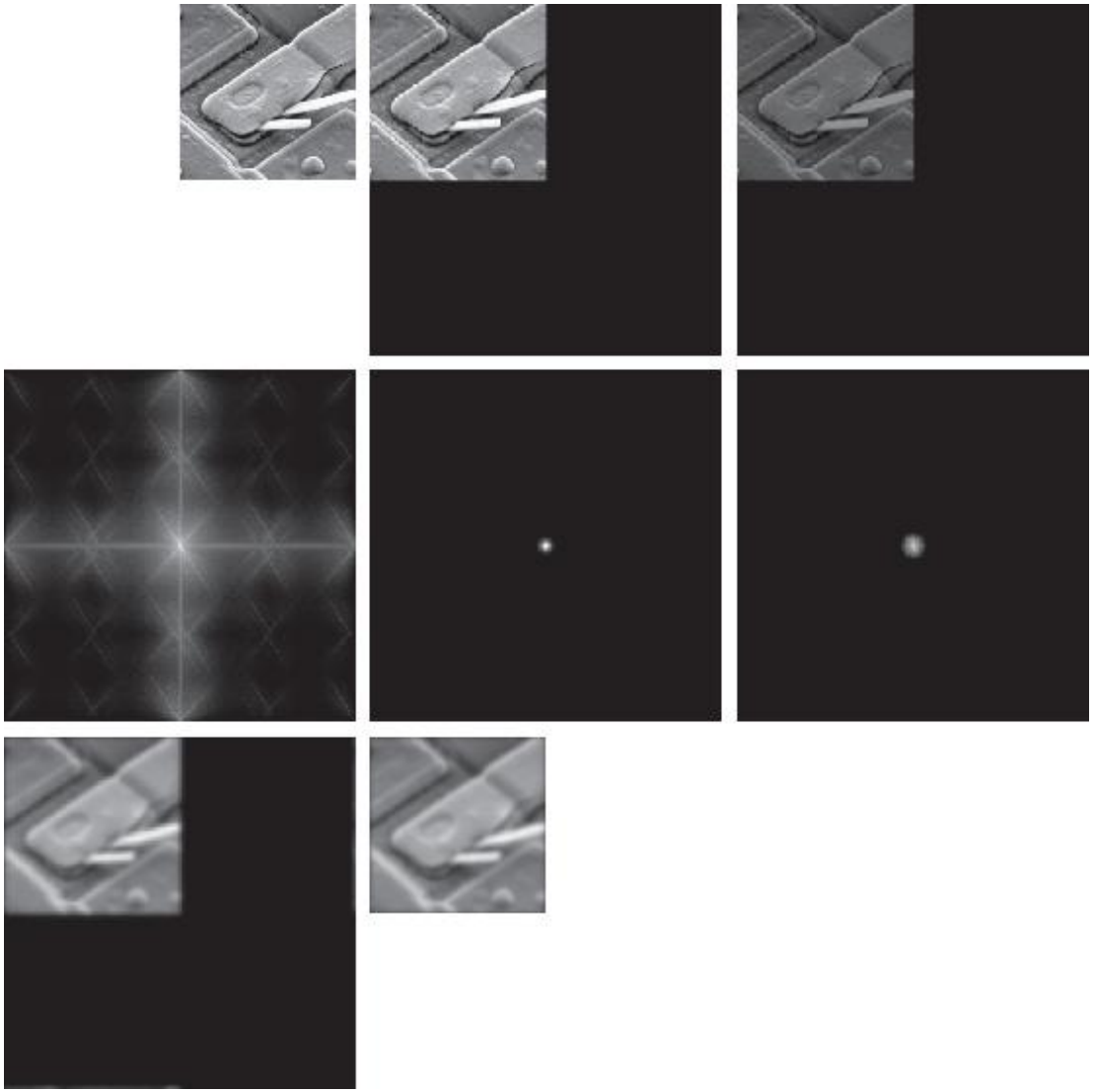
$$g_p(x,y) = \{real[\Im^{-1}[G(u,v)]]\}(-1)^{x+y}$$

7. 从步骤6中的结果$g_p(x,y)$的左上限提取$M \times N$区域，得到最终处理结果$g(x,y)$

Steps of filtering in the DFT domain

a b c
d e f
g h

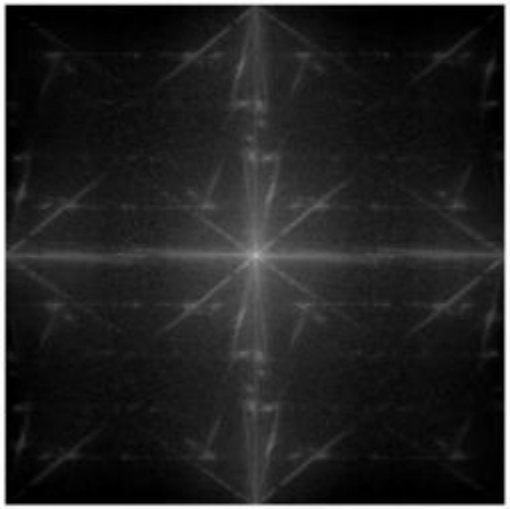
FIGURE 4.35
(a) An $M \times N$ image, f.
(b) Padded image, f_p of size $P \times Q$.
(c) Result of multiplying f_p by $(-1)^{x+y}$.
(d) Spectrum of F. (e) Centered Gaussian lowpass filter transfer function, H, of size $P \times Q$.
(f) Spectrum of the product HF.
(g) Image g_p, the real part of the IDFT of HF, multiplied by $(-1)^{x+y}$.
(h) Final result, g, obtained by extracting the first M rows and N columns of g_p.



Steps of filtering in the DFT domain

- What if the filter is known in the spatial domain?





-1	0	1
-2	0	2
-1	0	1

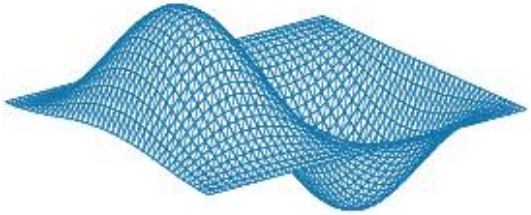
- Apply a 3x3 Sobel filter to the 600x600 image in the frequency domain.

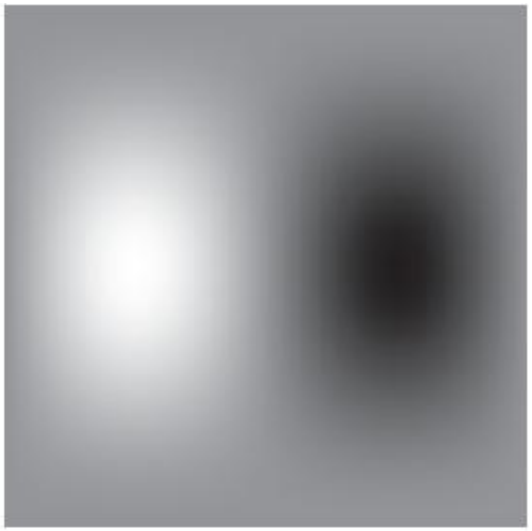
The importance of zero padding

a b
c d

FIGURE 4.38
(a) A spatial kernel and perspective plot of its corresponding frequency domain filter transfer function.
(b) Transfer function shown as an image.
(c) Result of filtering Fig. 4.37(a) in the frequency domain with the transfer function in (b).
(d) Result of filtering the same image in the spatial domain with the kernel in (a). The results are identical.

−1	0	1
−2	0	2
−1	0	1









Fast Fourier Transform

- The reason that Fourier based techniques have become so popular is the development of the **Fast Fourier Transform (FFT)** algorithm.
- It allows the Fourier transform to be carried out in a reasonable amount of time.
- Reduces the complexity from $O(N^4)$ to $O(N^2 \log N^2)$.

Frequency Domain Filtering & Spatial Domain Filtering

- Similar jobs can be done in the spatial and frequency domains.
- Filtering in the spatial domain can be easier to understand.
- Filtering in the frequency domain can be much faster – especially for large images.



Smoothing Frequency Domain Filters

- Smoothing is achieved in the frequency domain by dropping out the high frequency components
- The basic model for filtering is:

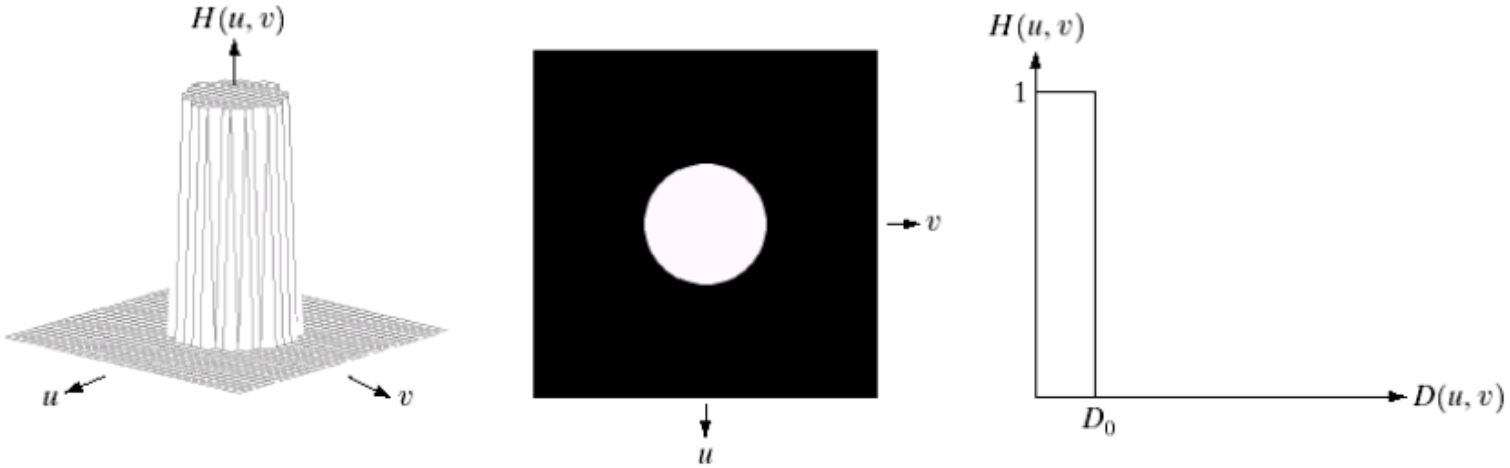
$$G(u,v) = H(u,v)F(u,v)$$

where $F(u,v)$ is the Fourier transform of the image being filtered and $H(u,v)$ is the filter transform function

- Low pass filters – only pass the low frequencies, drop the high ones.

Ideal Low Pass Filter

- Simply cut off all high frequency components that are a specified distance D_0 from the origin of the transform.



Changing the distance changes the behavior of the filter.

Ideal Low Pass Filter

- The transfer function for the ideal low pass filter can be given as:

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

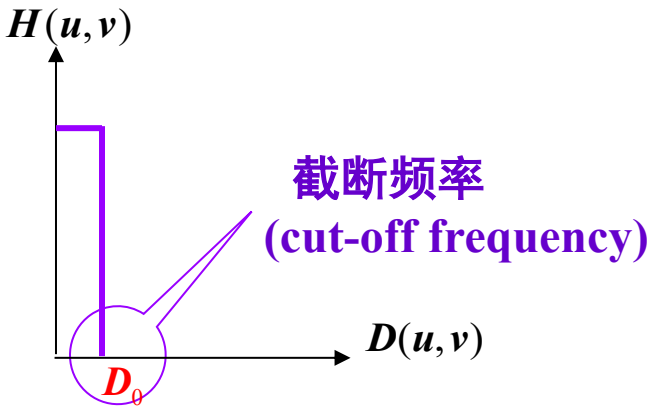
where $D(u,v)$ is given as:

$$D(u,v) = \left[\left(u - \frac{M}{2} \right)^2 + \left(v - \frac{N}{2} \right)^2 \right]^{1/2}$$

Ideal Low Pass Filter

How to choose cut-off frequency D_0?

$$H(u,v)=\begin{cases}1 & if\ D(u,v)\le D_0\\ 0 & if\ D(u,v)>D_0\end{cases}$$



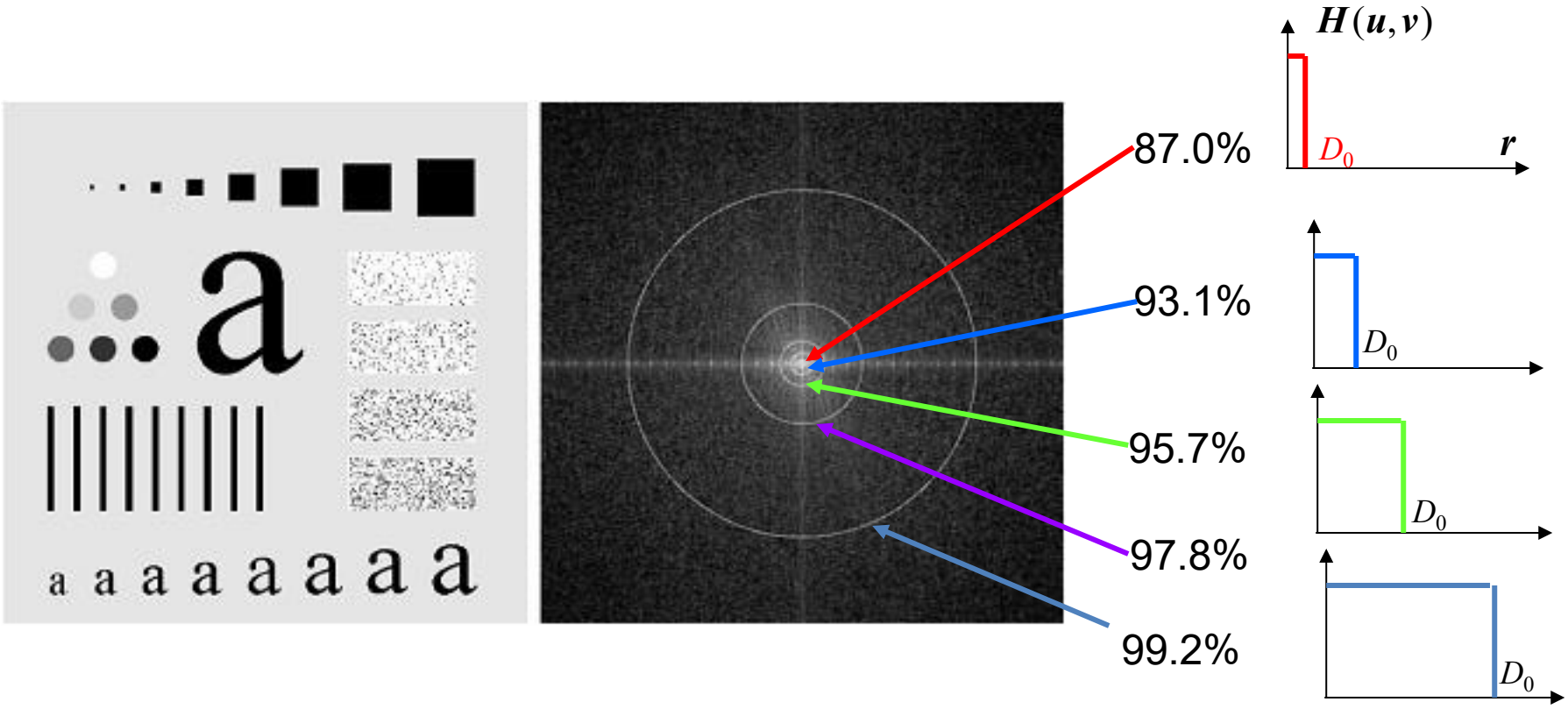
Use Fourier Power spectrum! (Remove the high frequency components)

低频能量所占比重: $\alpha=\dfrac{\text{图像区域能量}}{\text{图像所有能量}}=\dfrac{\sum_{r=0}^{D_0}\sum_{\theta=0}^{2\pi}|F(r,\theta)|^2}{\sum_{r=0}^{D_{\max}}\sum_{\theta=0}^{2\pi}|F(r,\theta)|^2}\times 100$

高频能量:

例如选取α=95%

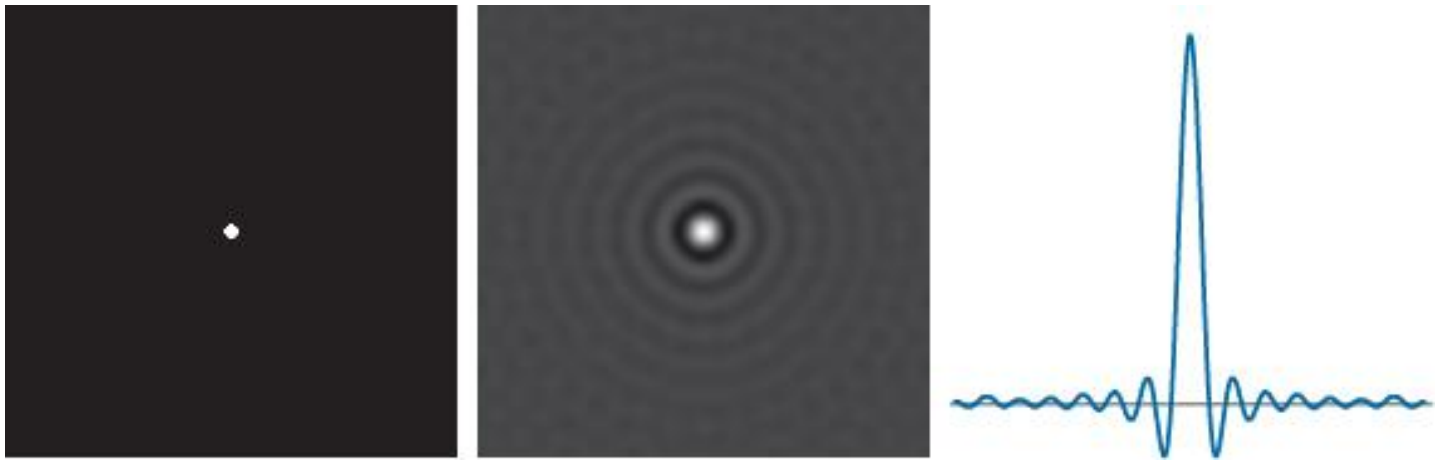
Ideal Low Pass Filter



An image, its Fourier spectrum and a series of ideal low pass filters of radius 10, 30, 60, 160 and 460 superimposed on top of it.

Ideal Low Pass Filter

- ILPF in the spatial domain is a *sinc* function that has to be truncated and produces ringing effects.
- The main lobe is responsible for blurring and the side lobes are responsible for ringing.



Ideal Low Pass Filter



1-100%=0%



1-99.2%=0.8%



1-95.7%=4.3%



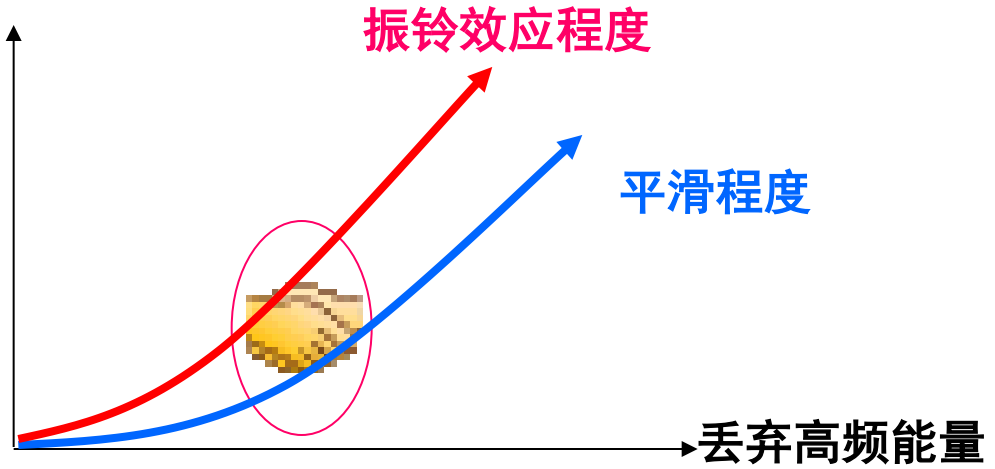
1-93.1%=6.9%



1-87%=13%



平滑程度和振铃(环)效应的矛盾

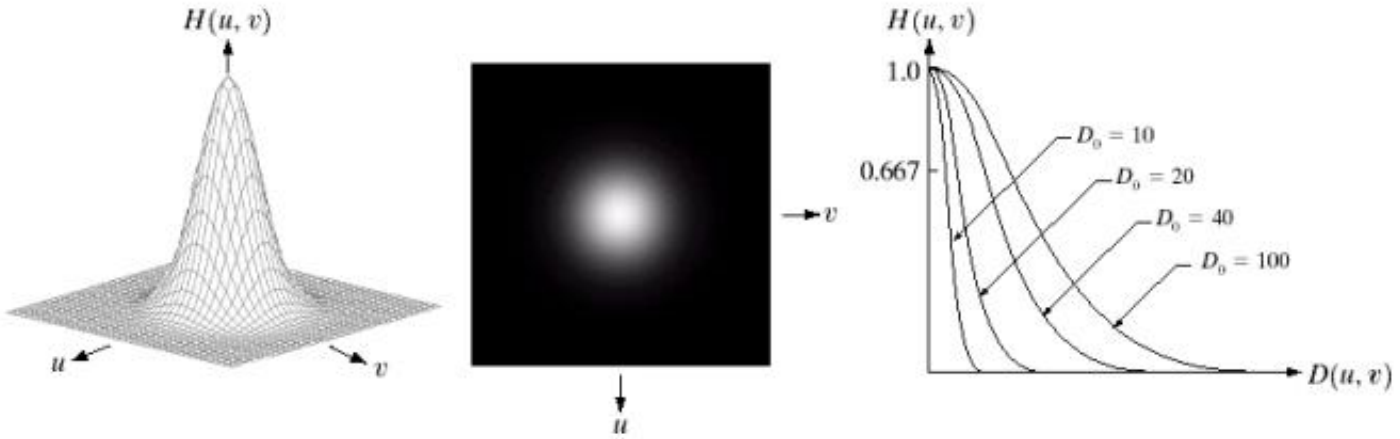


The filtered image can have negative values, so scaling normally is required

Gaussian Lowpass Filters

- The transfer function of a Gaussian lowpass filter is defined as

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$

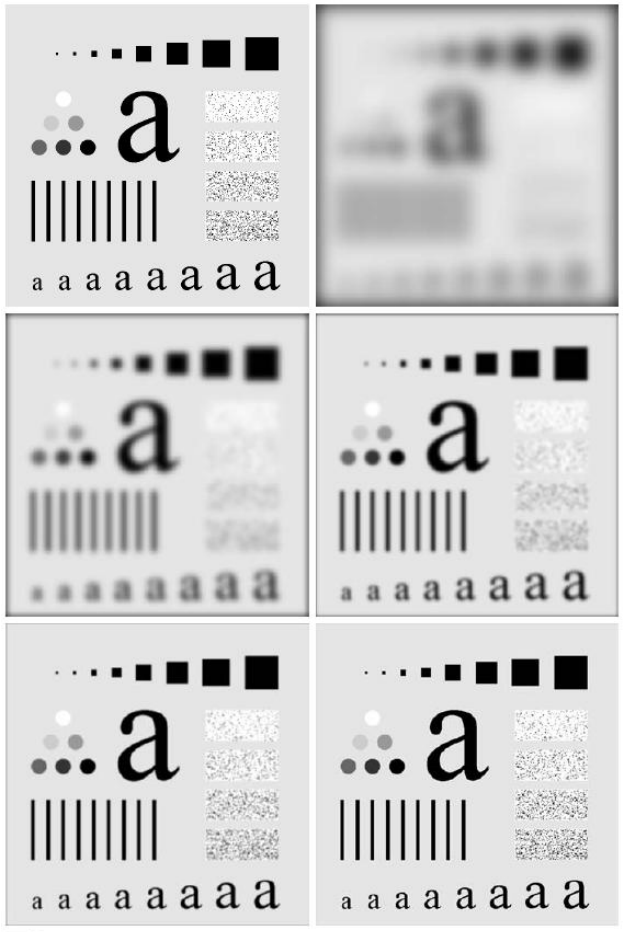


The inverse Fourier transform is Gaussian also. So Gaussian filters can avoid ringing

Gaussian Lowpass Filters

Example 1: Gaussian lowpass filtering

$$H(u,v) = e^{\frac{-D^2(u,v)}{2\sigma^2}}$$



No ringing at all

The filtered image can have no negative values

Gaussian Lowpass Filters

- **Example 2: Gaussian lowpass filtering**

$$H(u,v) = e^{\frac{-D^2(u,v)}{2\sigma^2}}$$

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Gaussian Lowpass Filters

- **Example 3: Gaussian lowpass filtering**

$$H(u,v) = e^{\frac{-D^2(u,v)}{2\sigma^2}}$$



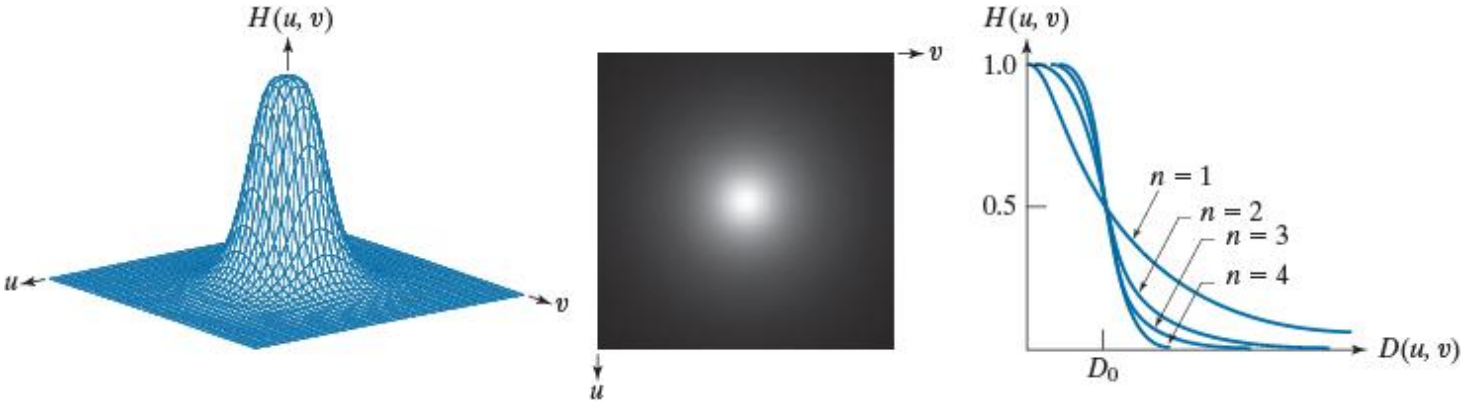
a b c

FIGURE 4.50 (a) Original image (784 × 732 pixels). (b) Result of filtering using a GLPF with $D_0 = 100$. (c) Result of filtering using a GLPF with $D_0 = 80$. Note the reduction in fine skin lines in the magnified sections in (b) and (c).

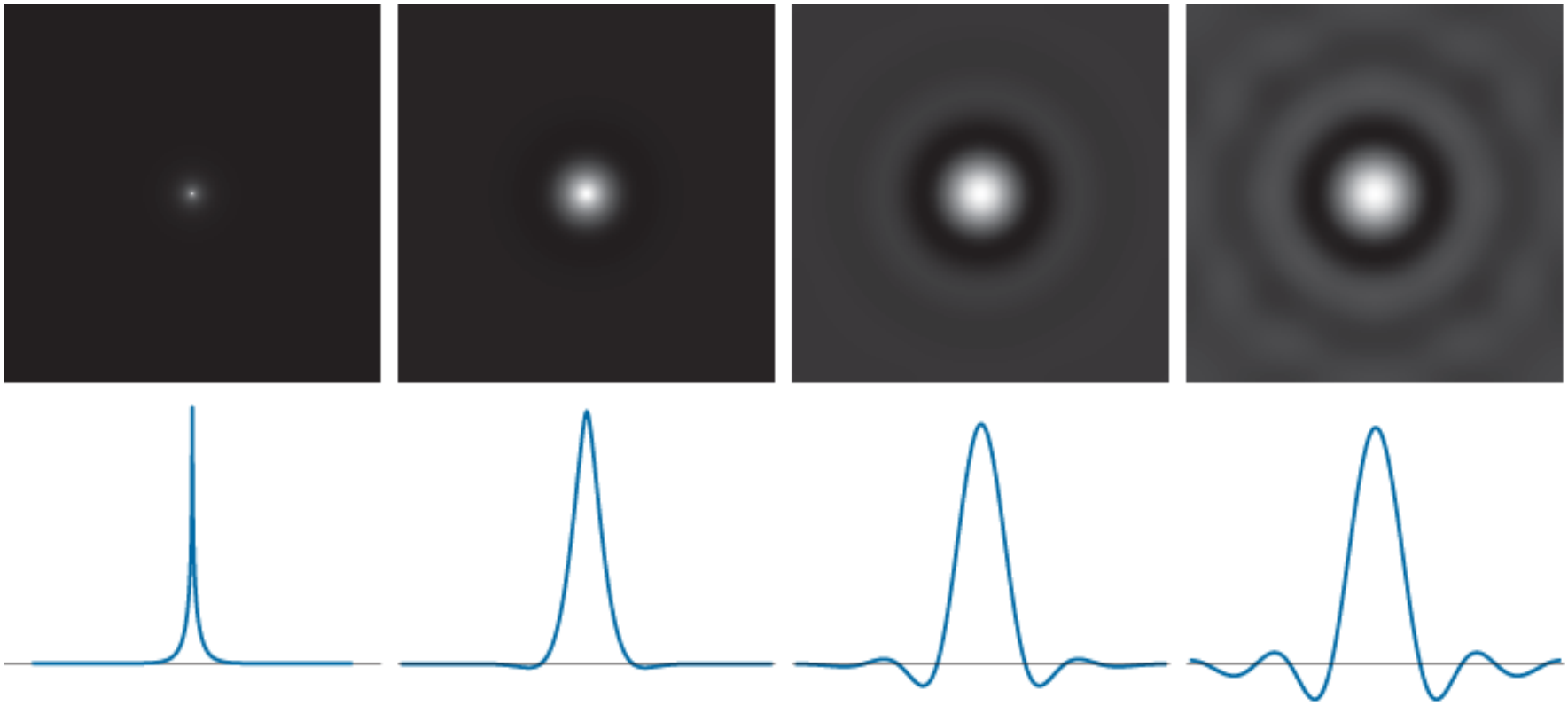
Butterworth Lowpass Filters

- The transfer function of a Butterworth lowpass filter of order n with cutoff frequency at distance D_0 from the origin is defined as:

$$H(u,v) = \frac{1}{1+\left[D(u,v)/D_0\right]^{2n}}$$



Butterworth Lowpass Filters



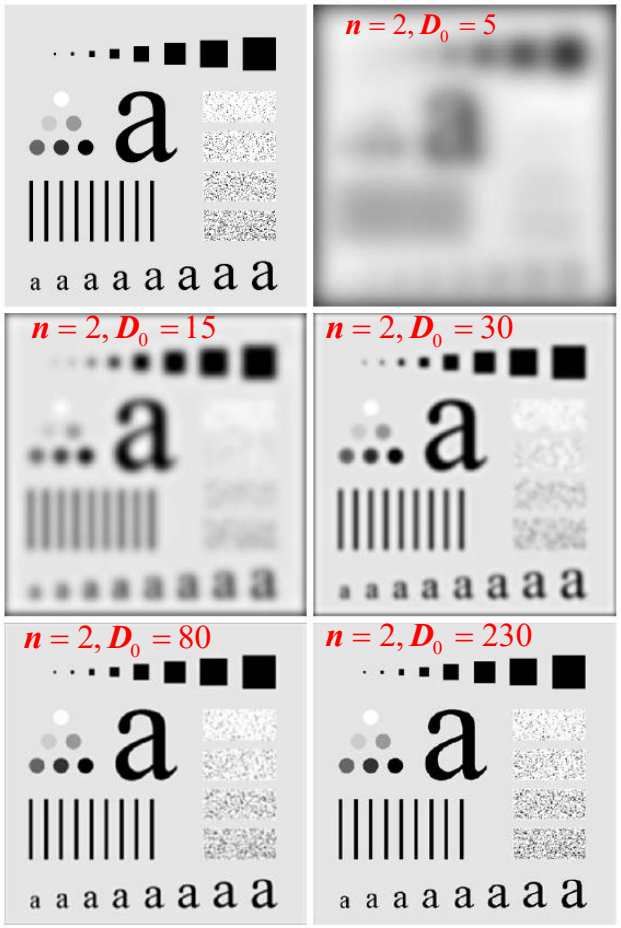
n=1 *n*=2 *n*=5 *n*=20

a b c d
e f g h

FIGURE 4.47 (a)–(d) Spatial representations (i.e., spatial kernels) corresponding to BLPF transfer functions of 1000×1000 pixels, cut-off frequency of 5, and order 1, 2, 5, and 20, respectively. (e)–(h) Corresponding inten profiles through the center of the filter functions.

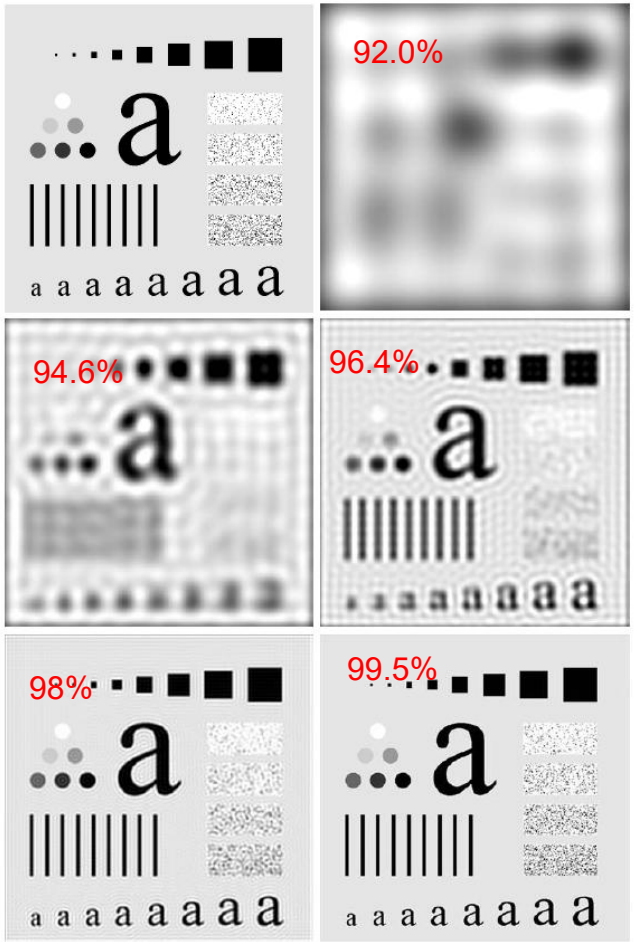
Butterworth Lowpass Filters

Butterworth Lowpass Filters



No ringing is visible

Ideal Low Pass Filter

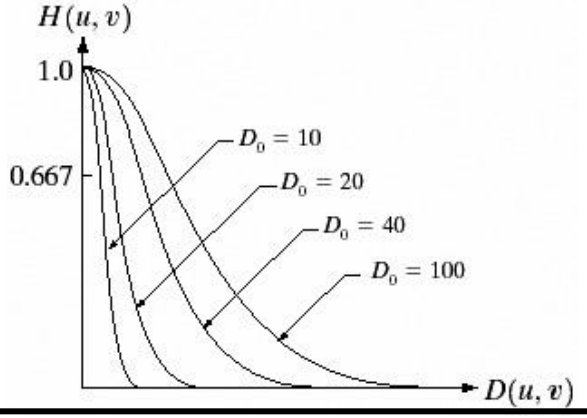


平滑和振铃之间难以取得平衡

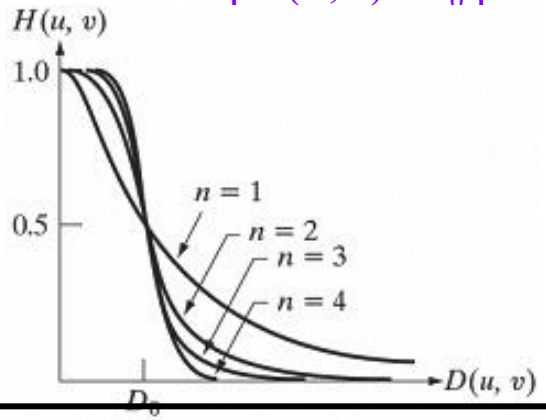
Gaussian Lowpass Filters

➢Gaussian lowpass vs. Butterworth lowpass

$$H(u,v) = e^{\frac{-D^2(u,v)}{2\sigma^2}}$$



$$H(u,v) = \frac{1}{1+\left[D(u,v)/D_0\right]^{2n}}$$



① **Gaussian can completely eliminate ringing while Butter worth cannot.**

② **At the cutoff frequency, Gaussian is not as sharp as Butterworth**

Butterworth order controls transition steepness. Higher orders may increase ringing.

Gaussian smoothing avoids oscillatory sidelobes. Choose parameters for the task.



Sharpening in the Frequency Domain

- Edges and fine detail in images are associated with high frequency components
- High pass filters – only pass the high frequencies, drop the low ones
- High pass frequencies are precisely the reverse of low pass filters, so:

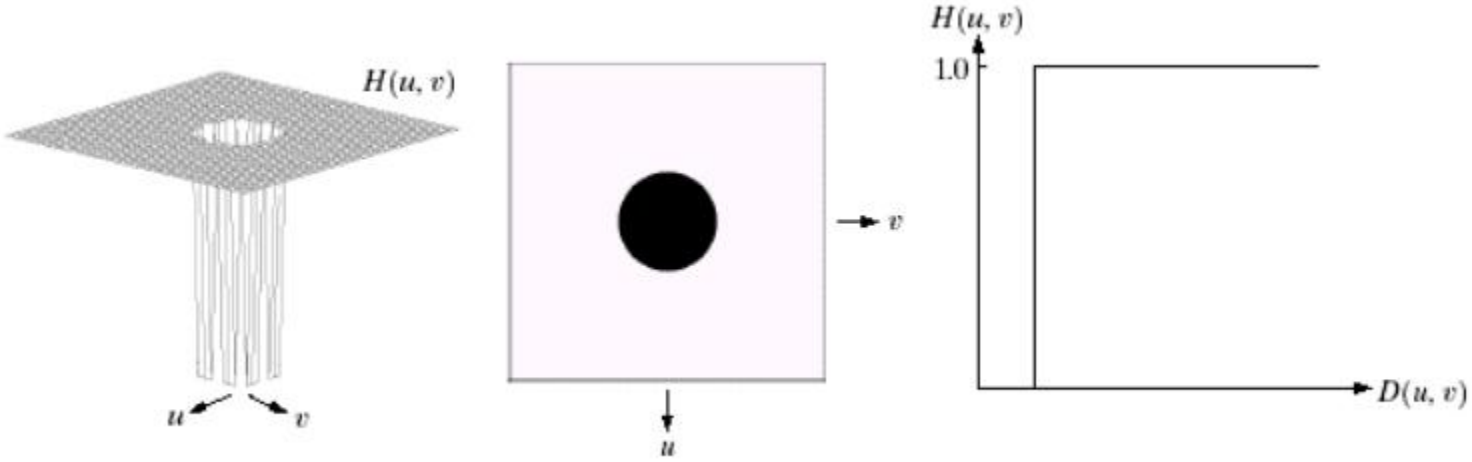
$$H_{hp}(u,v) = 1 - H_{lp}(u,v)$$

Ideal High Pass Filters

- The ideal high pass filter is given by:

$$H_{hp}(u,v) = \begin{cases} 0 & \text{if } D(u,v) \le D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

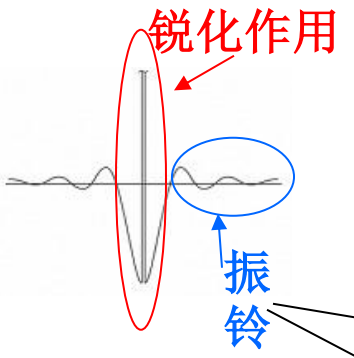
- D0 is the cut off distance as before.



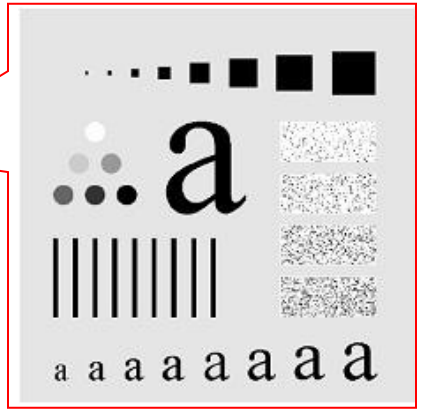
Ideal High Pass Filters

Example of ideal highpass filters

理想目标是:平坦区域全部变为黑色，变化部分变为白色



Input image



D_0越小，振铃效果越明显



a b c

cutoff截止频率
D_0=30

截止频率
D_0=60

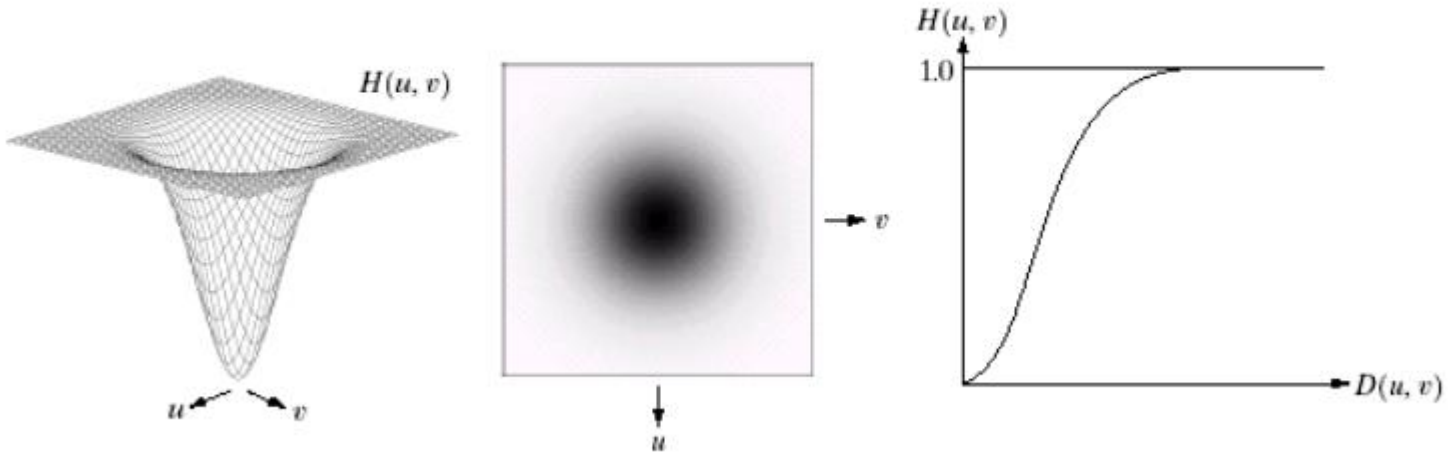
截止频率
D_0=160

Gaussian High Pass Filters

- The Gaussian high pass filter is given as:

$$H(u,v) = 1 - e^{\frac{-D^2(u,v)}{2D_0^2}}$$

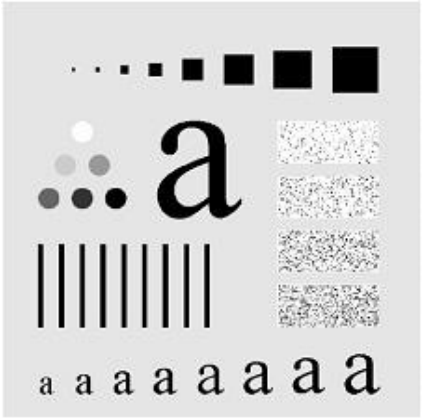
D_0 is the cut off distance as before.



Gaussian High Pass Filters

➢ Example of Gaussian highpass filtering

Input image



No oscillatory ringing



$D_0 = 30$ $D_0 = 60$ $D_0 = 160$

Butterworth High Pass Filters

- The Butterworth high pass filter is given as:

$$D(u,v) = \left[\left(u - \frac{M}{2}\right)^2 + \left(v - \frac{N}{2}\right)^2\right]^{1/2}$$

$$H(u,v) = \frac{1}{1 + \left[D_0 / D(u,v)\right]^{2n}}$$

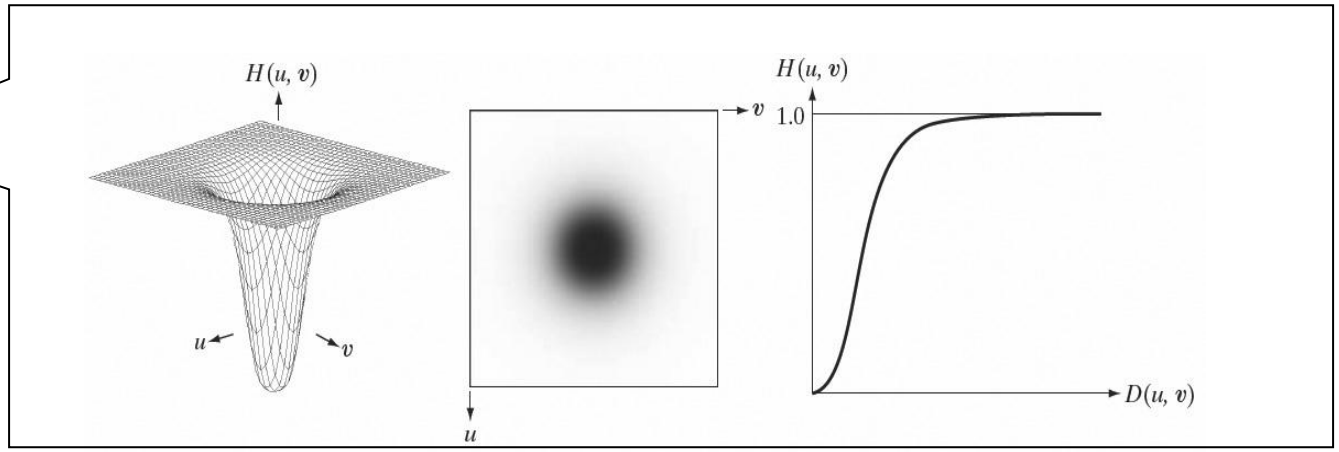
- **Transfer function of Butterworth lowpass filters:**

$$H(u,v) = \frac{1}{1 + \left[D(u,v) / D_0\right]^{2n}}$$

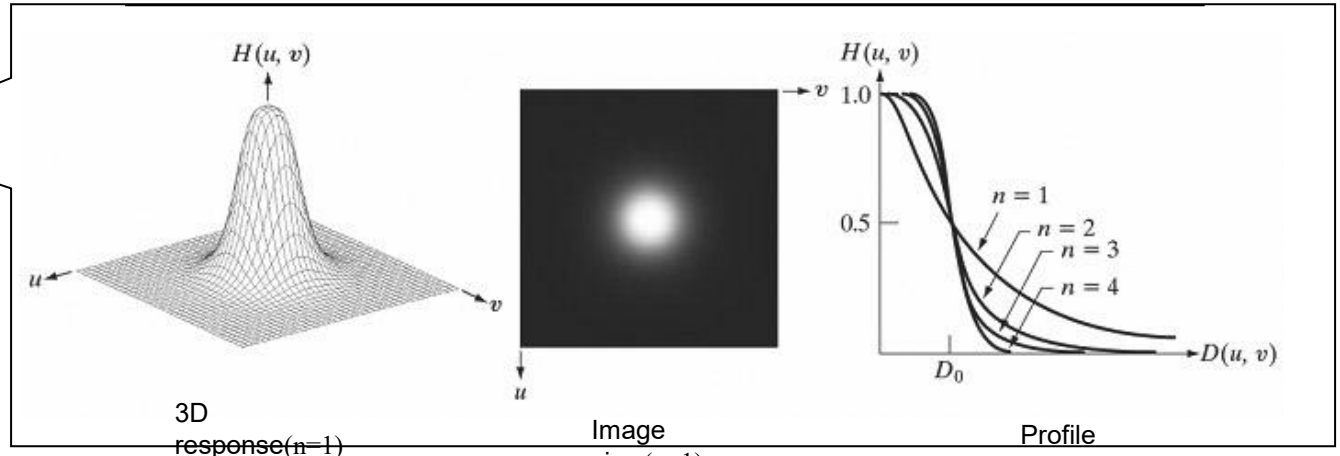
Butterworth High Pass Filters

Illustration of Butterworth high pass filters

Highpass



Lowpass



3D response(n=1)

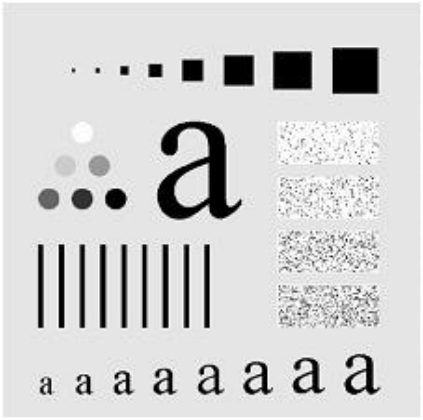
Image view(n=1)

Profile

Butterworth High Pass Filters

Example of Butterworh highpass filtering

原图像



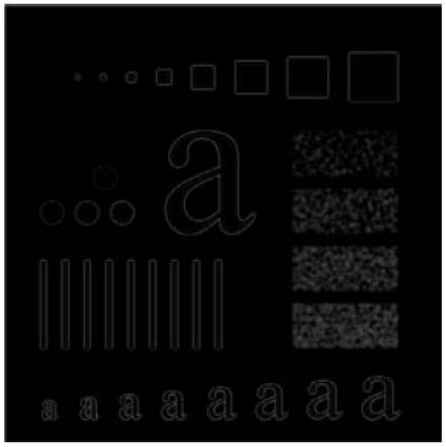
几乎没有ringing



$n = 2, D_0 = 30$



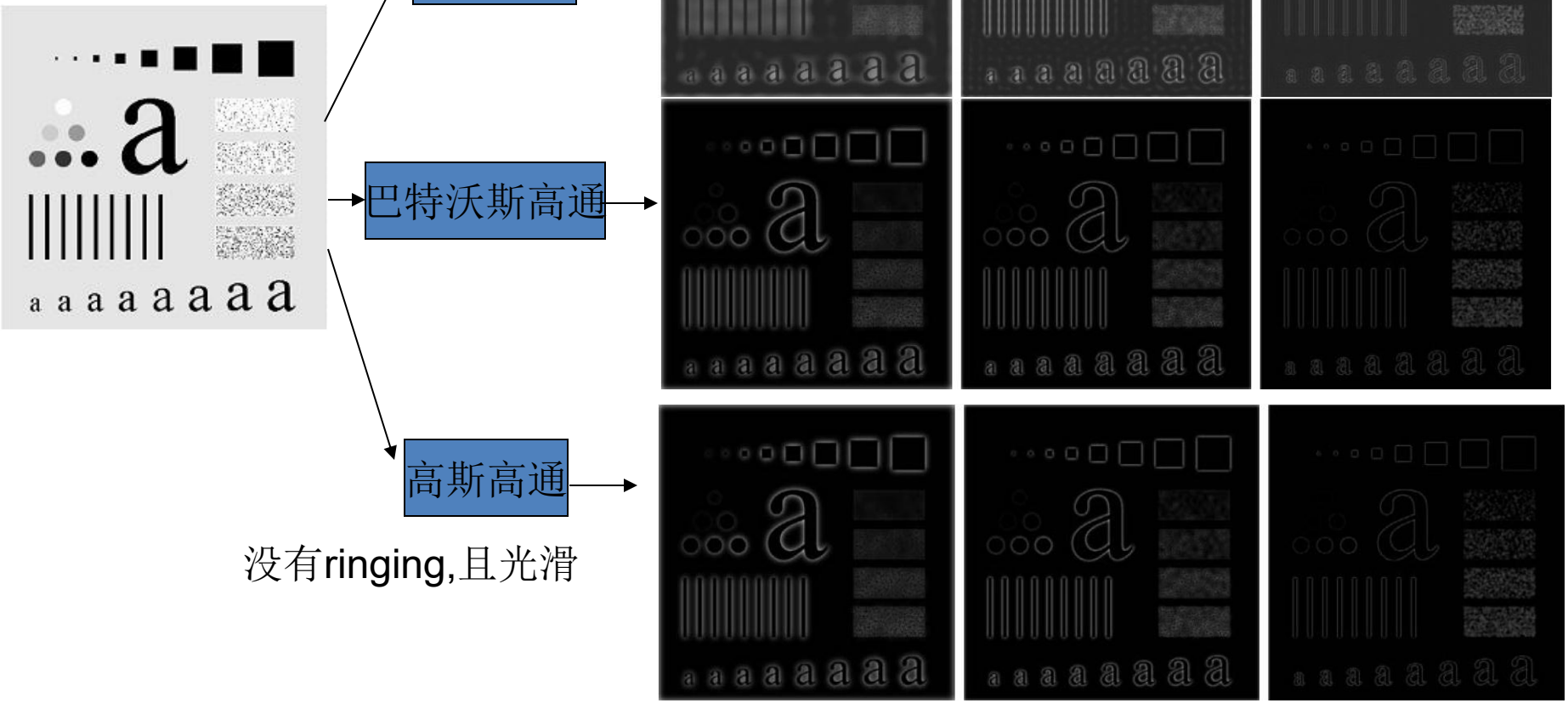
$n = 2, D_0 = 80$



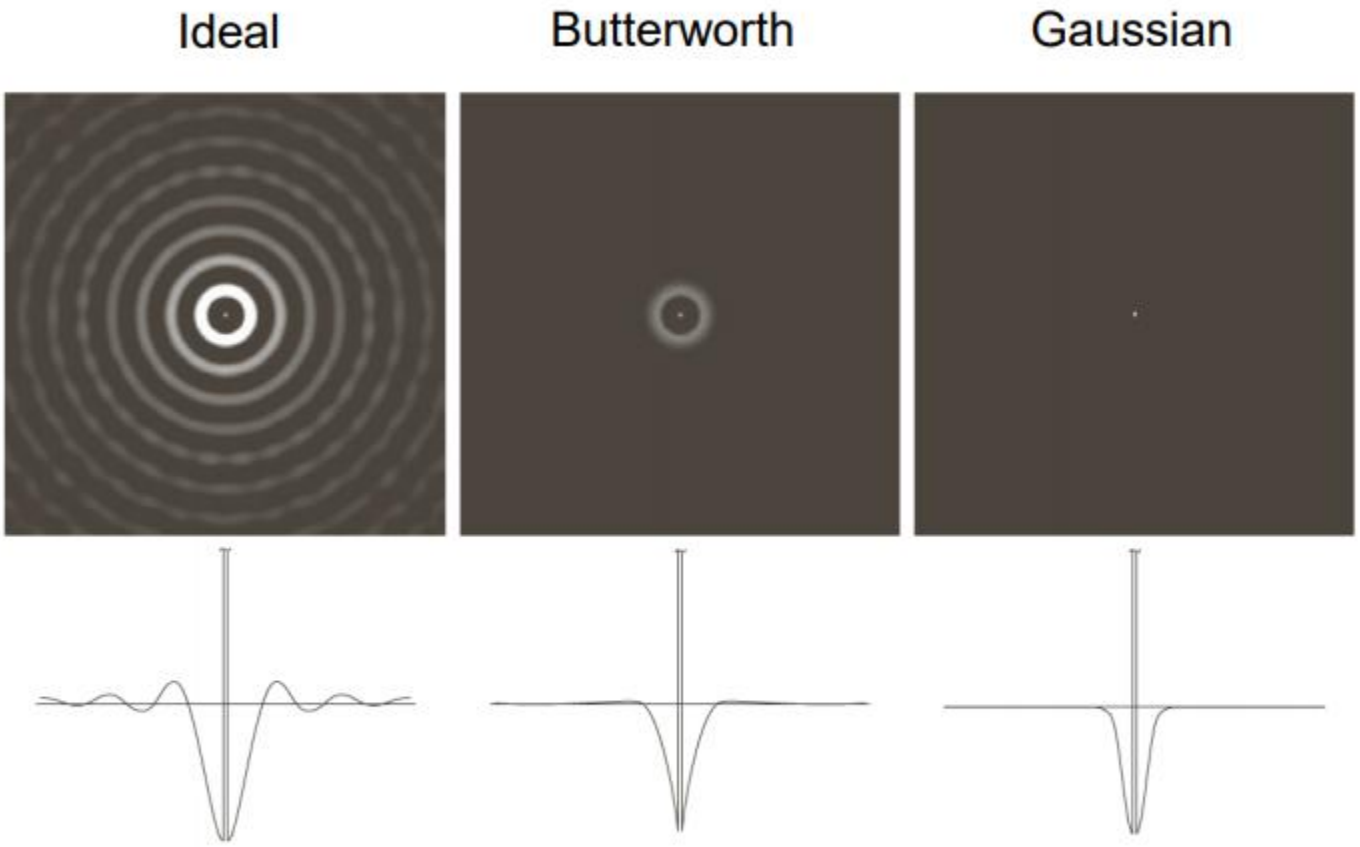
$n = 2, D_0 = 160$

Highpass Filter Comparison

➢**Ideal vs. Butterworth vs. Gaussian highpass**



Spatial Representation of Highpass Filters



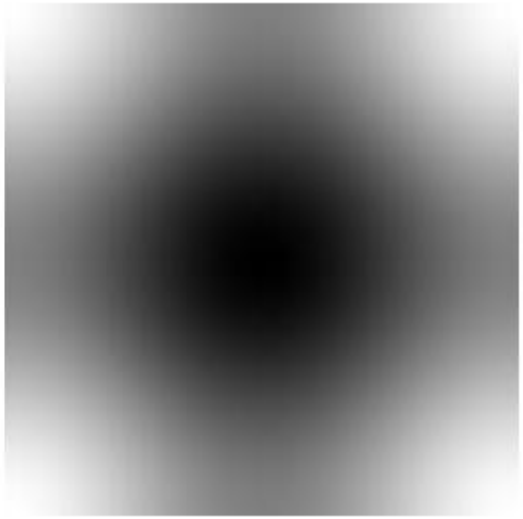
a b c

FIGURE 4.53 Spatial representation of typical (a) ideal, (b) Butterworth, and (c) Gaussian frequency domain highpass filters, and corresponding intensity profiles through their centers.

The Laplacian in the Frequency Domain

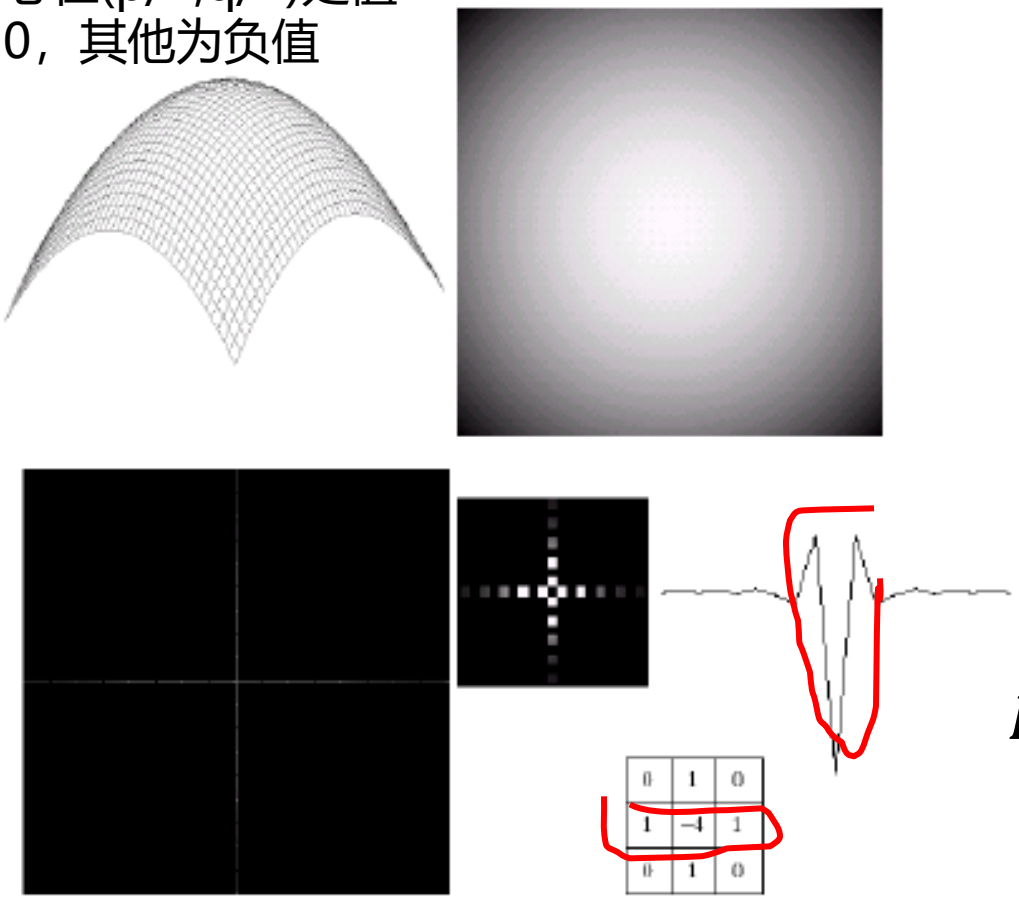
- Image enhancement operations (e.g. unsharp masking, high boost filtering) may be alternatively implemented in the frequency domain.
- Laplacian in the DFT domain:

$$H(u,v) = -4\pi^2(u^2 + v^2)$$



The Laplacian in the Frequency Domain

中心在(p/2,q/2)处值为0，其他为负值



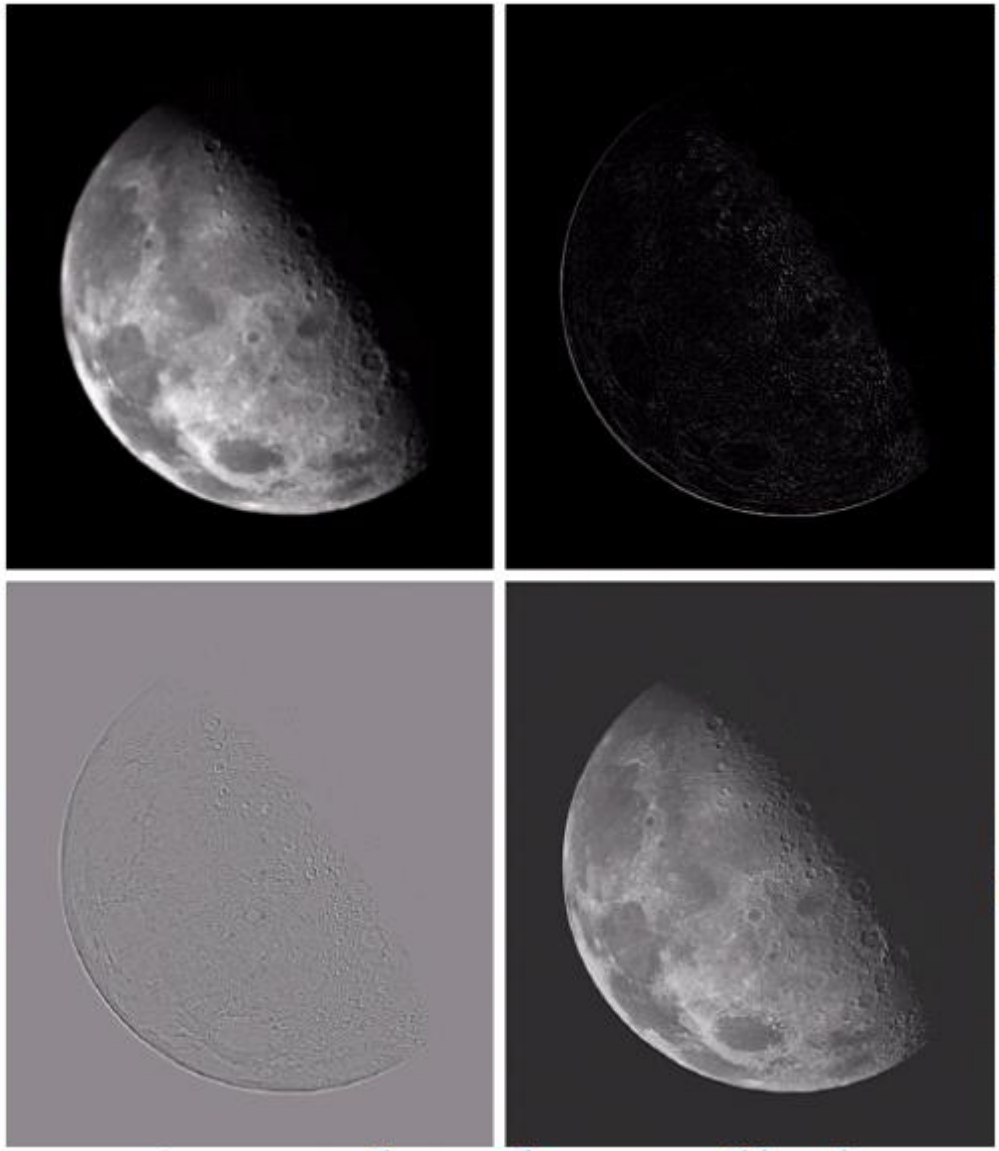
a) 拉普拉斯算子滤波器在频域的三维图形
b) 滤波器的频谱图像
c) 通过傅里叶逆变换在空域的图像
d) 空域图象的中心部分
e) 围绕中心点的灰度值分布
f) 空域拉普拉斯模板

$$H(u,v) = -4\pi^2 D^2(u,v)$$

a b c d e f

FIGURE 4.27 (a) 3-D plot of Laplacian in the frequency domain. (b) Image representation of (a). (c) Laplacian in the spatial domain obtained from the inverse DFT of (b). (d) Zoomed section of the origin of (c). (e) Gray-level profile through the center of (d). (f) Laplacian mask used in Section 3.7.

Frequency Domain Laplacian Example



Frequency Domain Laplacian Example

Original Image



Frequency Domain Laplacian



Spatial Domain Laplacian



Unsharp Masking, Highboost Filtering and High-frequency-Emphasis Filtering

？为什么要进行高频提升和高频加强？

所有高通滤波后的图象都有一个共同特点：背景的平均强度减小到接近黑色；

这是因为高通滤波器除去了傅立叶变换的零频率成分 $(F(0,0)=0)$

- 解决办法：把原始图像加入到滤波后的结果中去，如拉普拉斯算子增强，这种处理称为高频提升过滤。它是钝化模板的推广。

Unsharp Masking, Highboost Filtering and High-frequency-Emphasis Filtering

- 钝化模版：从一幅图像减去其自身模糊图像从而生产锐化图像。

在频率域，即从图像本身减去低通滤波（模糊）后的图像而得到高通滤波（锐化）的图像

$$g_{mask}(x,y) = f(x,y) - f_{LP}(x,y)$$
$$f_{LP}(x,y) = \mathfrak{F}^{-1}\{H_{LP}(x,y)F(u,v)\}$$

- 钝化模版和高频提升过滤

$$g(x,y) = f(x,y) + kg_{mask}(x,y)$$
$$= \mathfrak{F}^{-1}\{[1 + k * [1 - H_{LP}(x,y)]]F(u,v)\}$$
$$g(x,y) = \mathfrak{F}^{-1}\{[1 + k * H_{HP}(u,v)]F(u,v)\}$$

当K=1时，即钝化模板；当k>1，高提升滤波

Unsharp Masking, Highboost Filtering and High-frequency-Emphasis Filtering

$$g(x,y) = \mathfrak{F}^{-1}\{[1 + k * H_{HP}(u,v)]F(u,v)\}$$

其中，红色部分称为高频强调滤波器，高通滤波器通过将直流项置为0，导致滤波后图像的平混灰度减小为0。而高频强调滤波器可避免这一问题。

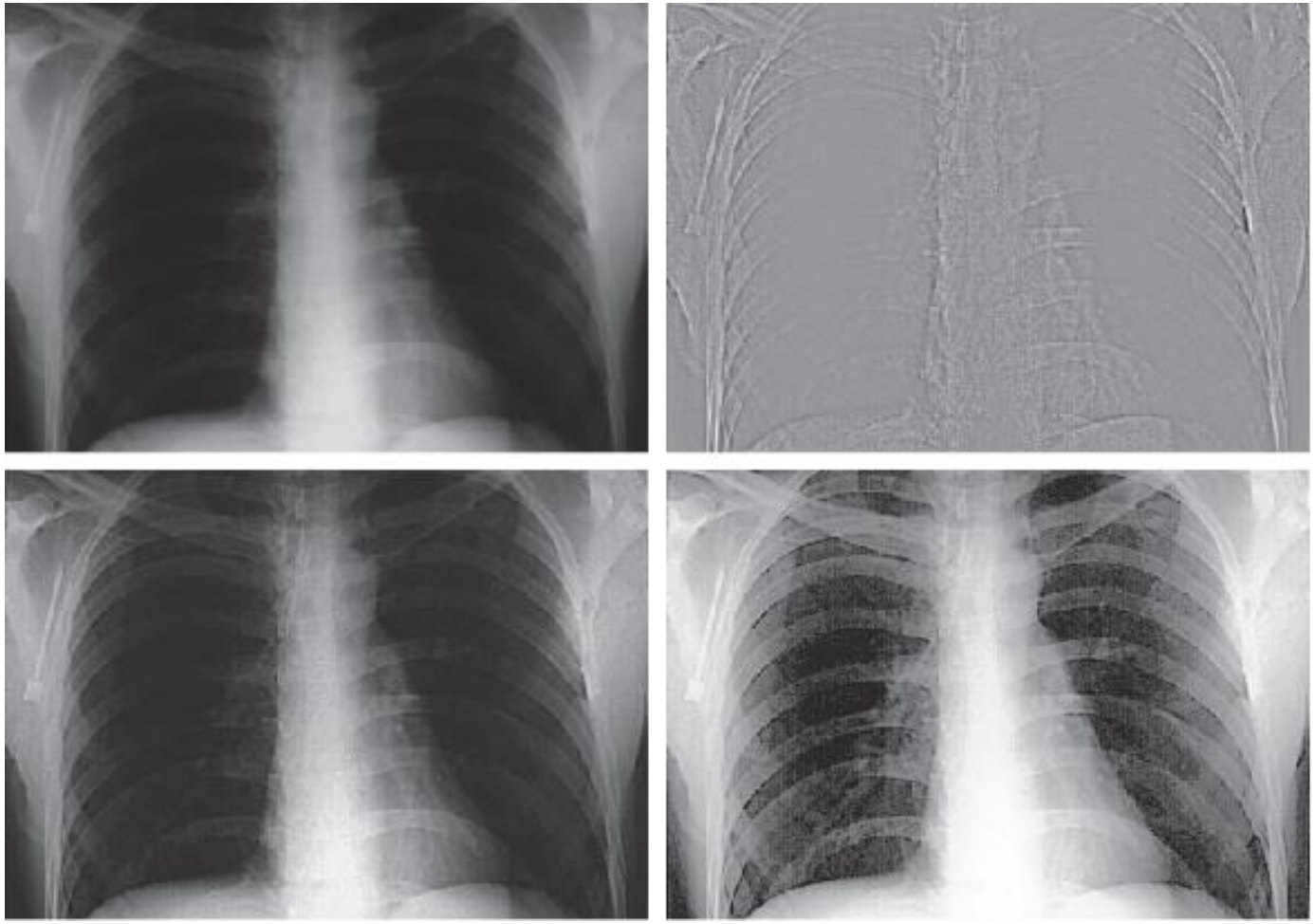
“1”：高通滤波器加1，使得零频率不被滤波器滤除

“K”：决定了影响最终结果的高频的比例

更一般的公式为：

$$g(x,y) = \mathfrak{F}^{-1}\{[k_1 + k_2 * H_{HP}(u,v)]F(u,v)\}$$

Unsharp Masking, Highboost Filtering and High-frequency-Emphasis Filtering



a b
c d

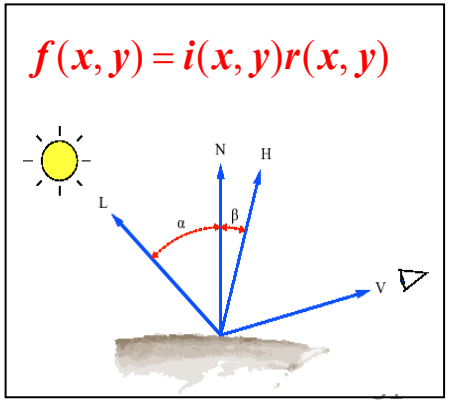
FIGURE 4.57
(a) A chest X-ray. (b) Result of filtering with a GHPF function. (c) Result of high-frequency-emphasis filtering using the same GHPF. (d) Result of performing histogram equalization on (c). (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

Homomorphic Filter

- The illumination-reflectance can be used to develop a frequency domain procedure for improving the appearance of an image by simultaneous intensity range compression and contrast enhancement.

$$f(x,y) = i(x,y)r(x,y)$$

- The illumination component of an image generally is characterized by slow spatial variations, while the reflectance component tends to vary abruptly, particularly at the junctions of dissimilar objects.
  - Illumination components $\Longrightarrow$ Low frequency
  - Reflectance components $\Longrightarrow$ High frequency



Homomorphic Filter

➢**Log transform separates the illumination and reflectance components**

Before Transform

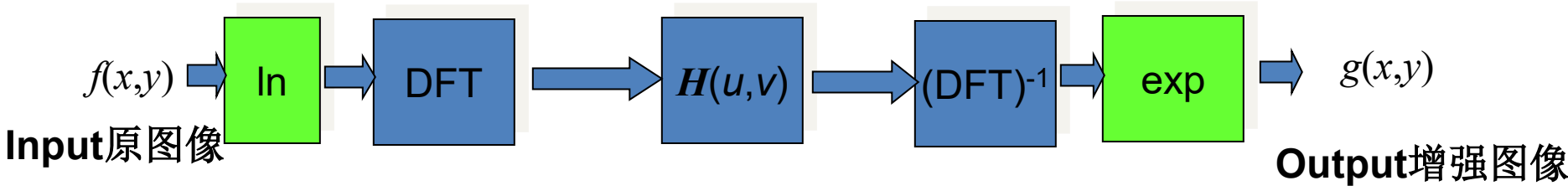
$$\Im\{f(x,y)\} = \Im\{i(x,y)r(x,y)\} \neq \begin{cases} \Im\{i(x,y)\}\Im\{r(x,y)\} \\ \Im\{i(x,y)\} + \Im\{r(x,y)\} \end{cases}$$

$$z(x,y) = \ln[f(x,y)] = \ln[i(x,y)] + \ln[r(x,y)]$$

after Transform

$$\Im\{z(x,y)\} = \Im\{\ln[f(x,y)]\} = \Im\{\ln[i(x,y)]\} + \Im\{\ln[r(x,y)]\}$$

➢Homomorphic Filter flowchart

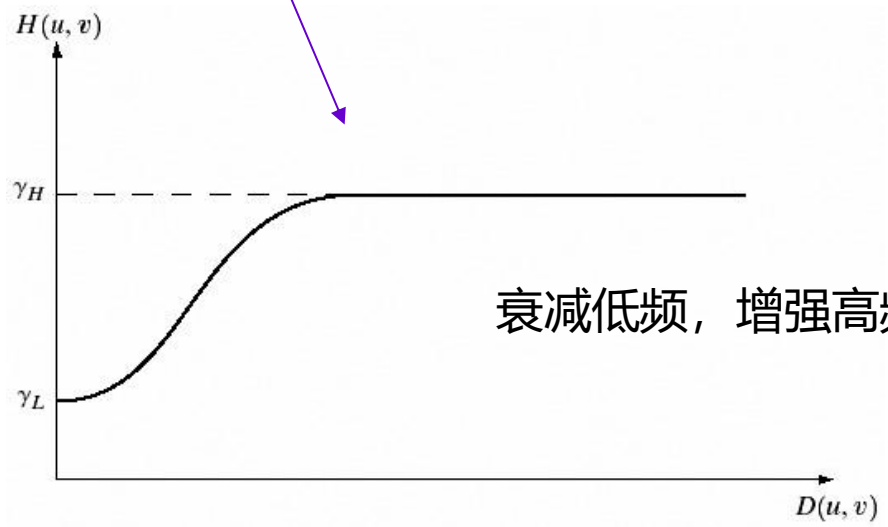


Homomorphic Filter

- The homomorphic filter is given as:

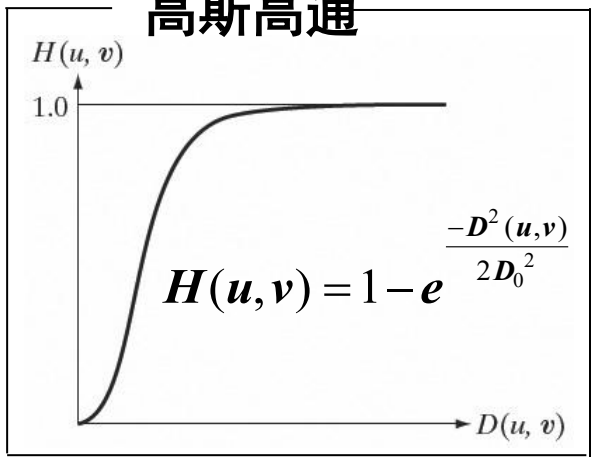
$$H(u,v) = (r_H - r_L)\left[1 - e^{-c\frac{D^2(u,v)}{D_0^2}}\right] + r_L$$

c控制函数的偏斜角



衰减低频，增强高频

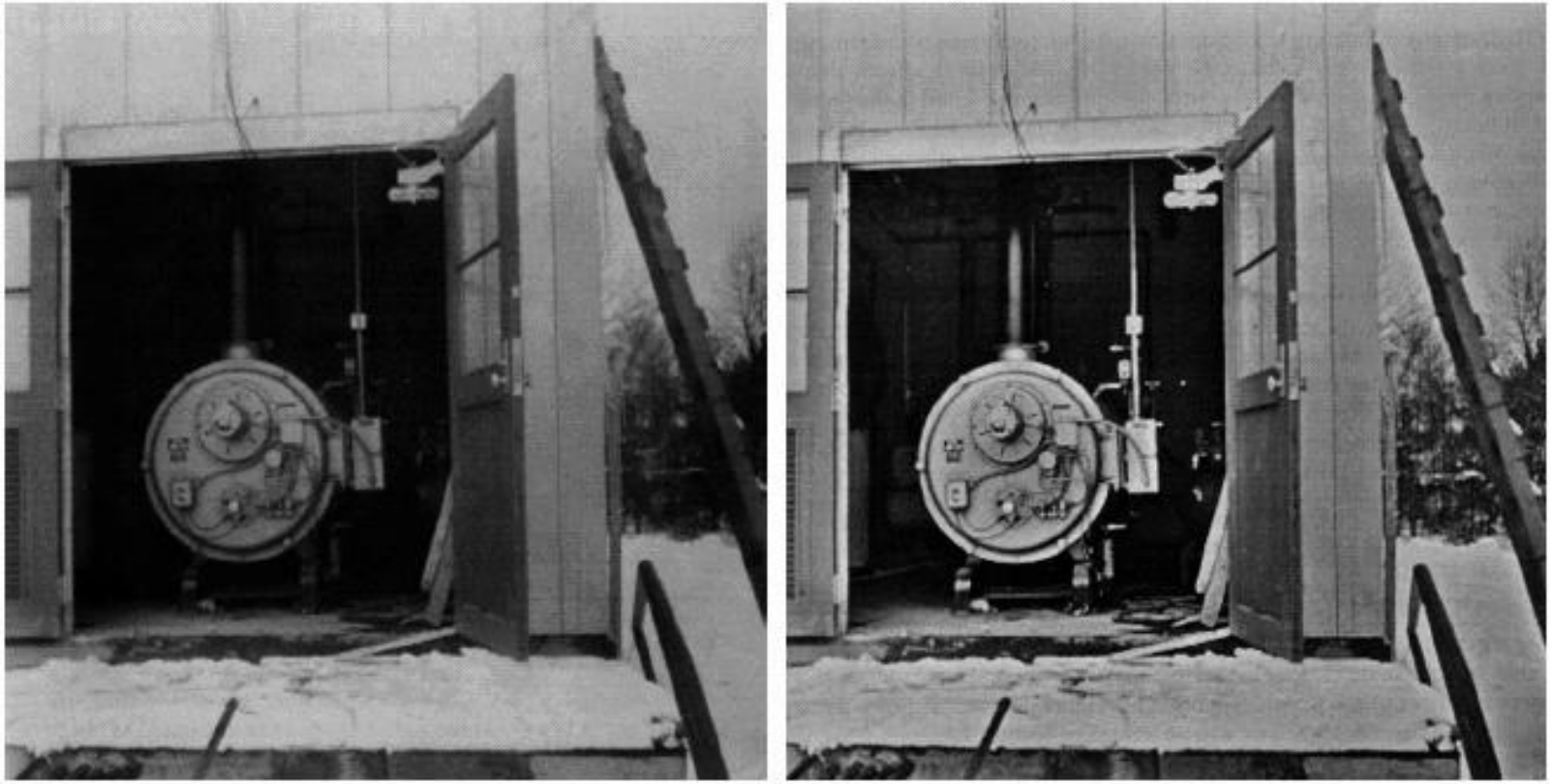
高斯高通



滤波器的径向剖面图

Homomorphic Filter

- Example



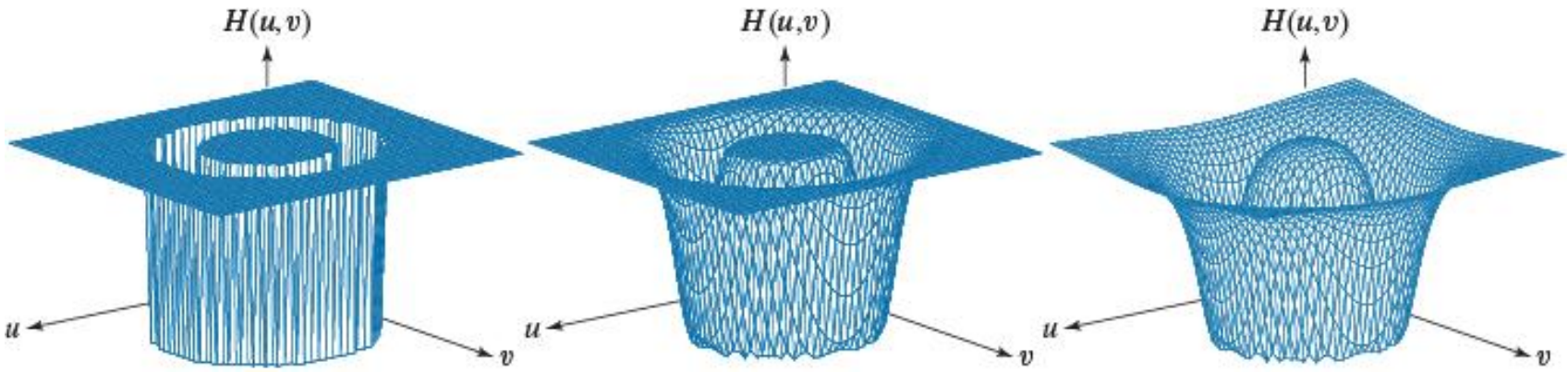
A reduction of dynamic range in the brightness, together with an increase in contrast, brought out the details of objects inside the shelter and balanced the gray levels of the outside wall. The enhanced image also is sharper.

Selective Filtering

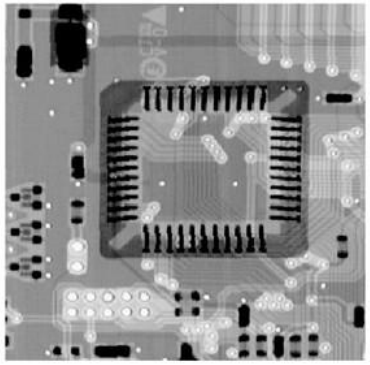
Bandreject and Bandpass Filters

TABLE 4.7
Bandreject filter transfer functions. C_0 is the center of the band, W is the width of the band, and $D(u,v)$ is the distance from the center of the transfer function to a point (u,v) in the frequency rectangle.

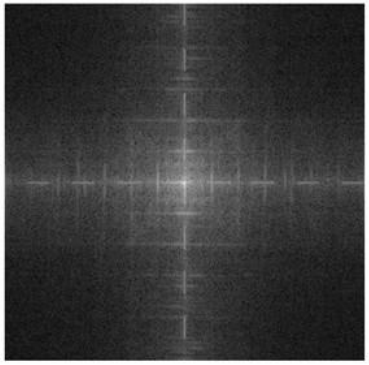
Ideal (IBRF)	Gaussian (GBRF)	Butterworth (BBRF)
$H(u,v) = \begin{cases} 0 & \text{if } C_0 - \frac{W}{2} \le D(u,v) \le C_0 + \frac{W}{2} \\ 1 & \text{otherwise} \end{cases}$	$H(u,v) = 1 - e^{-\left[\frac{D^2(u,v) - C_0^2}{D(u,v)W}\right]^2}$	$H(u,v) = \frac{1}{1 + \left[\frac{D(u,v)W}{D^2(u,v) - C_0^2}\right]^{2n}}$



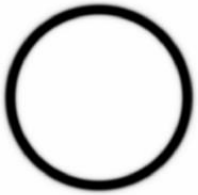
Selective Filtering

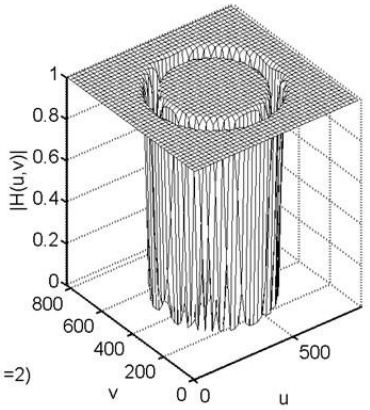


a).Original Image

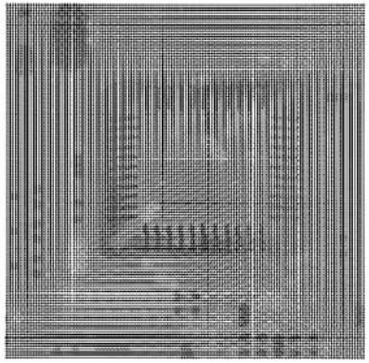


b).Fourier spectrum of a

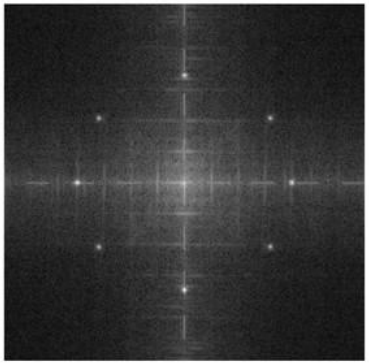




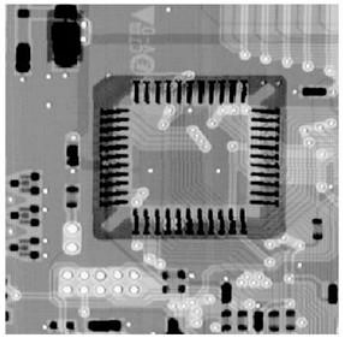
d).Butterworth Bandpass filter(D=355 W=40 n =2)



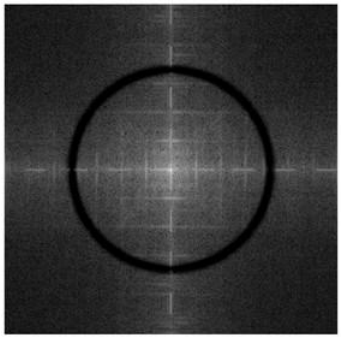
b).Result of add noise



c).Fourier spectrum of b

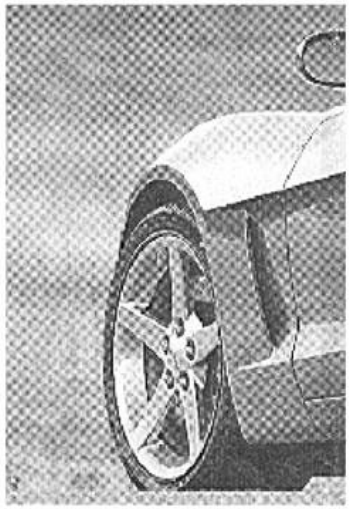


f).Result of denoise

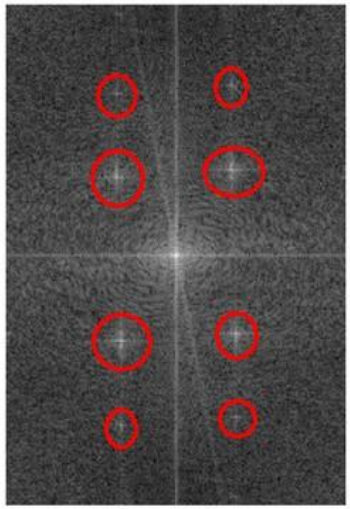


e).Fourier spectrum of f

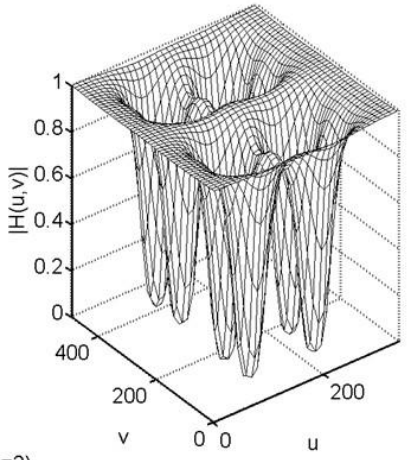
Selective Filtering



a).Original Image



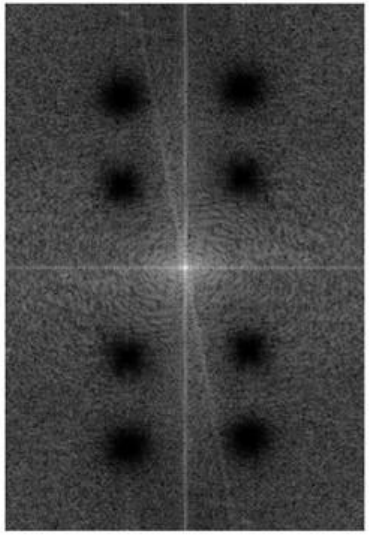




c).Butterworth Bandpass filter(D=355 W=40 n =2)

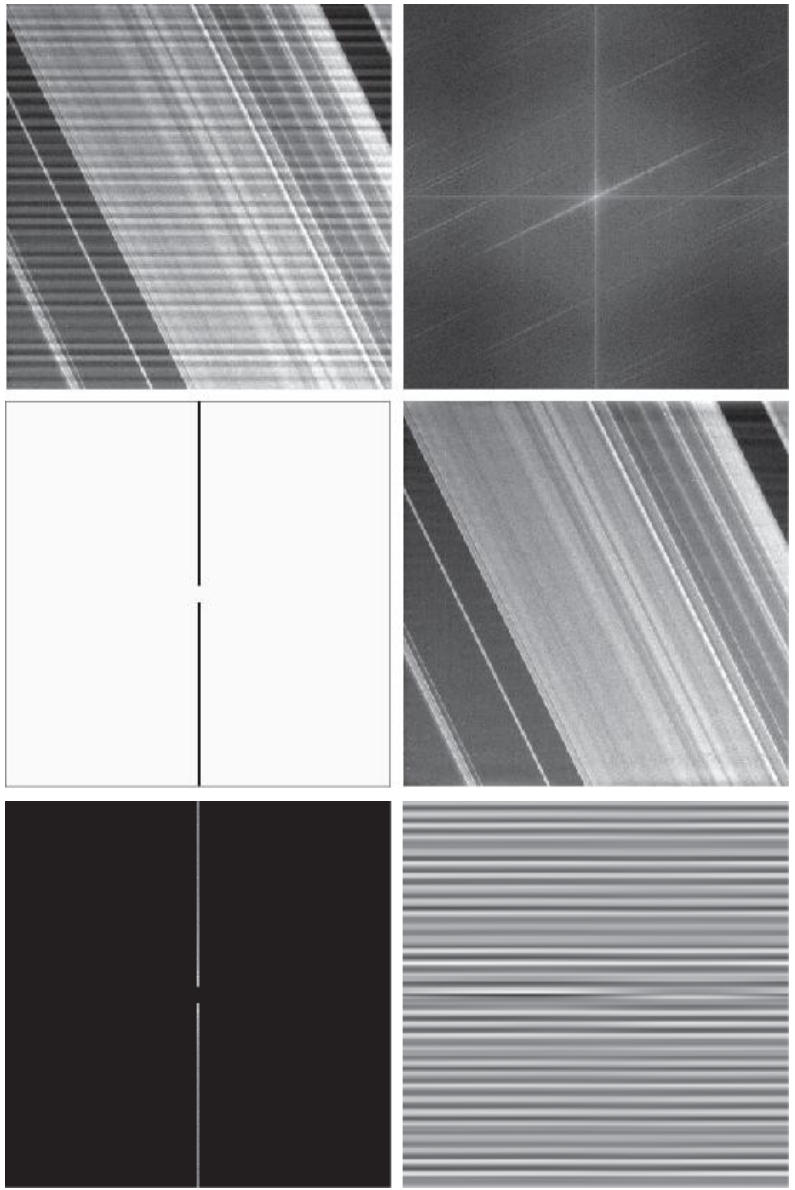


d).Result image



e).Result of filtering using d

Selective Filtering



a b
c d

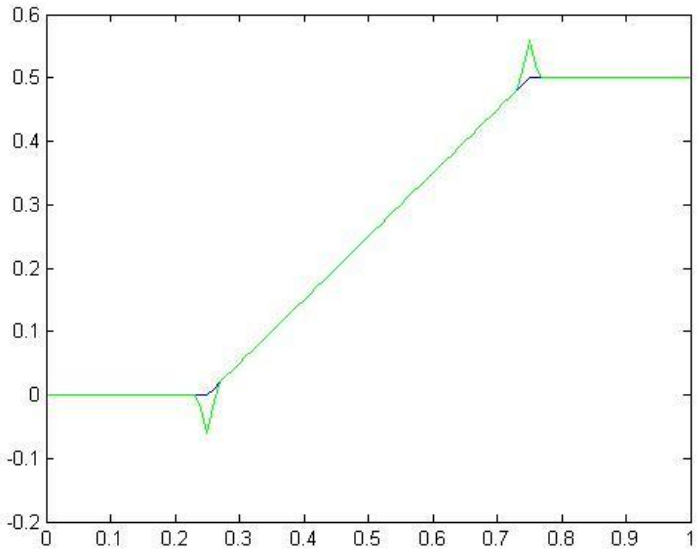
FIGURE 4.65
(a) Image of Saturn rings showing nearly periodic interference. (b) Spectrum. (The bursts of energy in the vertical axis near the origin correspond to the interference pattern). (c) A vertical notch reject filter transfer function. (d) Result of filtering. (The thin black border in (c) is not part of the data.) (Original image courtesy of Dr. Robert A. West, NASA/ JPL.)

Homework

1. 对于如下一维函数, 用unsharp masking进行处理并观察其边缘锐化结果。

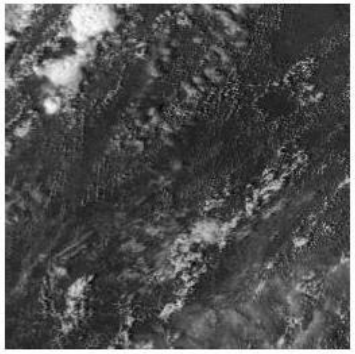
$$f(x)=\begin{cases}0, & 0\le x<0.25\\ x-0.25, & 0.25<x\le 0.75\\ 0.5, & 0.75\le x<1\end{cases}$$

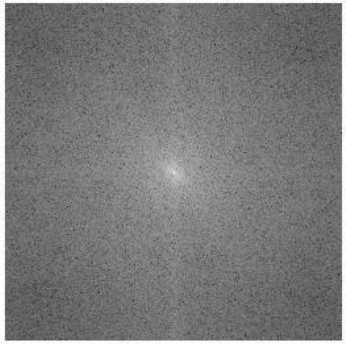
```
x=0:0.01:1;
f=0*(x>=0&x<=0.25)+(x-
0.25).*(x>0.25&x<=0.75)+0.5*(x>0.
75&x<=1); %一维函数f
plot(x,f);
axis([0,1,-0.2,0.6]);
g1=smooth(f);
mask=f-g1;
g2=f+10*mask; %unsharp masking
hold on;
plot(x,g2,'g');
```

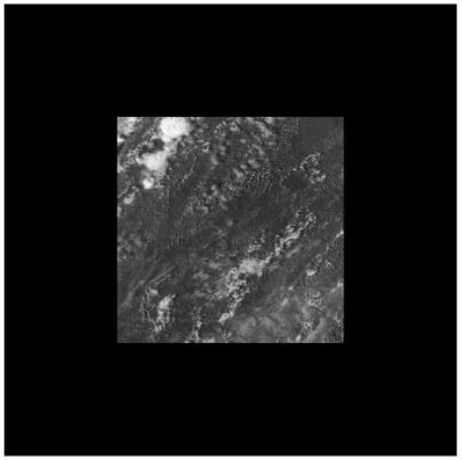


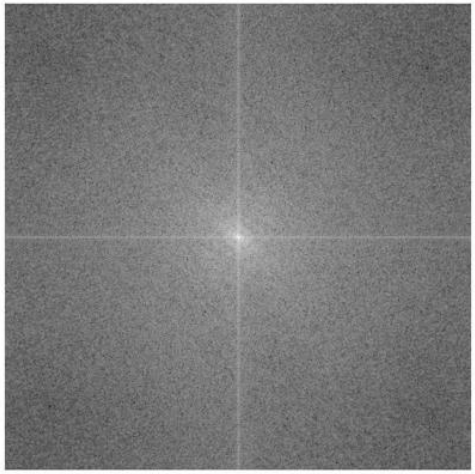
Homework

将一幅图像截取为大小是256x256的图像I，进行傅里叶变换。将I四周补零，进行傅里叶变换，并将两个结果进行比较，解释为什么补零后的在水平方向和竖直方向的分量会变强。









补0操作相当于给图像加上高频边框，增强了水平和竖直方向的周期性。故增强了水平和竖直方向的频谱。

频率域滤波器

- 频率域滤波基础
  - 补零
  - 频率域滤波的步骤小结
- 频率域滤波器平滑图像
  - 理想低通滤波器
  - 高斯低通滤波器
  - 巴特沃斯低通滤波器
- 频率域滤波器锐化图像
  - 理想、高斯和巴特沃斯高通滤波器
  - 频率域中的拉普拉斯
  - 钝化掩蔽、高提升滤波和高频强调滤波
  - 同态滤波
- 选择性滤波